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REPORT: RESEARCH PRACTICE

Modeling the dynamics of change in the System of Meanings Personal through the PsychLab Virtual Laboratory

Idiographic pilot study and research proposal on procrastination technology

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3. Context description

This Research Practicum has been developed within the framework of the Research Group Constructivist Research of the National University of Distance Education (GICUNED), directed by Dr. Luis Ángel Saúl Gutiérrez, professor at the Faculty of Psychology at UNED. The group is part of the constructivist paradigm. psychological and, specifically, in the tradition of Construct Psychology Personal accounts by George A. Kelly, whose development in the Spanish-speaking world has been driven by authors such as Botella and Feixas.

The group's activity is structured around the PsychLab virtual laboratory, a platform of research and intervention that integrates the classic idiographic techniques of the constructivist assessment—primarily the RepGrid Technique and the Weighted Implications Grid (WimpGrid)—with contemporary developments of Network Science and Fuzzy Logic. The central objective of this line of work, The field in which this report is framed is mathematical modeling and simulation of the dynamics of change in the system of personal meanings, which allows to graphically represent the cognitive structure of a subject, to identify the constructs nuclear issues and the implicative dilemmas that sustain resistance to change, and monitor objectively the evolution of therapeutic processes.

During the internship period, the group was developing several lines of simultaneous work, among which the evolution of the platform towards its new stand out version (PsychLab 2.0) and the PsychSim project, focused on interview simulation Therapeutic sessions with virtual patients using language models (LLMs). The students The Practicum student actively and collaboratively integrated into these lines, participating both in training and tool validation tasks and in project design of their own research.

The work modality was primarily telematic, typical of the nature of the work. distance from the UNED, which implied continuous coordination with the group of research through online meetings, *regular meeting spaces with the team teaching and collaborative document management tools (OneDrive, Word, Excel, Teams).* The Associated Center of reference has been Lugo, under the tutelage of the professor from the Headquarters, Dr. Luis Ángel Saúl Gutiérrez.

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4. Activities carried out

The activities carried out during the Practicum (December 15, 2025 – March 26)
(from 2026) have been developed within the framework of the PsychLab virtual laboratory and the Group of
Constructivist Research of the UNED (GICUNED), under the line of «Modeling
of dynamics of change in the Personal Meaning System." The set of tasks
It has been structured around four major blocks whose integration has given rise to the
results that are communicated in the Research Report.

4.1. Immersion and initial training (December 2025 – January 2026)

The incorporation into the research group began with attendance at the three conferences
Introduction to Practices and the PsychoCoffee sessions, meeting spaces of
team where progress, doubts and proposals regarding the lines of
open investigations.

These sessions proved especially valuable, both in terms of guidance and intellectual understanding.
The presentations by Dr. Luis Ángel Saúl and Alejandro Sanfeliciano allowed me to solidify...
realistic expectations about the scope and limitations of PsychLab, adjusting my
initial understanding of the laboratory and providing a more mature perspective on its
methodological possibilities. This initial anchor proved especially useful in relation to the
generation of ideas for the Final Degree Project, as it helped me to identify lines of
research consistent with the type of methodology available in the group.

On the other hand, the conference presented proposals that awakened in me a
genuine interest in the boundaries between constructivist psychology and the sciences
computational. In particular, I would like to highlight Dr. Gerardo Gómez's presentation on
reinforcement learning techniques and neural network training—a
This perspective, although far removed from the classic frameworks of TCP, raises questions
fascinating on the formal modeling of psychological change—, as well as the proposal
by Carlos Hurtado on indeterminacy and shared meanings from an approach
computational, based on the use of vector embeddings and deep architectures
learning to compare and quantify the degree of similarity between systems of meanings
personal accounts from different subjects. Both presentations confirmed for me that PsychLab
It operates in a genuinely interdisciplinary territory, in which the Psychology of the
Personal Constructs engages productively with the most formal tools
contemporary.

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In parallel, the technical environment required for the
laboratory data analysis work:

- Jamovi: Applied Statistics and Exploratory Analysis.
- RStudio: familiarization with the interface and some basic resources of the R programming environment.
- Markdown: reproducible documentation of procedures.

As an essential theoretical foundation for integration into PsychLab, the MOOC was completed. "Introduction to PsychLab", completed with the preparation of a critical report of review and feedback of the course itself (Appendix I), focusing on its didactic structure, content sequencing, platform usability, and concrete proposals for pedagogical improvement.

4.2. Specialized training and self-assessment with PsychLab (January – February 2026)

Next, the PsychLab Specialization Course was taken, structured in six
Topics that progressively cover the theoretical-applied framework of the laboratory:

Topic	Content
1	Introduction to Network Theory in Psychology of Change
2	Evaluation of Personal Meanings with the Grid Technique (RepGrid)
3	Modeling the Personal Meanings System (WimpGrid)
4	Identifying psychological obstacles ("knots") in change processes
5	Case formulation with Fuzzy Cognitive Maps (FCM)
6	Monitoring the change process using mathematical indices

Each topic involved developing a reasoned reflection in the forum and applying it
Practice of the concepts on clinical cases and graphic simulations.

The final assessment activity for the course consisted of a complete self-assessment. through PsychLab, structured around the core construct of self-demand and its Contingency with academic stress. This work, which forms part of the body of the Report

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Research included self-characterization, elicitation of personal constructs,
RepGrid and WimpGrid administration, network topology analysis, and
Comparison of pre/post-intervention measures (Appendix II).

4.3. Design of the group project on procrastination (February – March) 2026)

Alongside the specialized training, there was active participation —along with the rest
of the Practicum students— in the design of a quasi-research project
experimental study on "Technological procrastination, personality and construct system"
personal tasks." Collective tasks included bibliographic research and review.
(academic procrastination, Cloninger's psychobiological model, Fear of Missing Out e
digital hygiene), the operational delimitation of variables, the methodological integration of
psychometric instruments (TCI-R-67, EPA) with the constructivist methodology
(Self-characterization and WimpGrid), the digitization of the questionnaires and the development
of the project appendices (Appendix III). The complete project was submitted on the 26th
March 2026 as the final milestone of the Practicum.

Within the team, my work focused specifically on the neurobiological dimension of
phenomenon—the role of dopamine and neurotransmitters in the circuits of
rewards involved in technological procrastination—. On a technical level,
I developed an automation procedure using Google Apps Script for
Transfer the TCI-R-67 scale (originally in PDF format) to a Google Form
Forms with the format fully adapted, significantly streamlining their administration
telematics and reducing the margin of error in data collection. With a view to the
Continuing the investigation, he would also assume the tasks of statistical analysis of the
data, applying the correlational and repeated measures tests provided for in the plan
analysis.

4.4. Complementary (cross-cutting) participation

As cross-cutting activities that have enriched practical training,
They developed three additional lines.

4.4.1. PsychSim testing and prompt development

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We collaborated in the testing phase of PsychSim, a tool of the PsychLab 2.0 project that It uses language models (LLMs) to simulate therapeutic interviews with patients virtual. Participation resulted in the complete design of two system prompts:

- **Dr. JSON (patient generator):** an agent specializing in generating patient profiles fictitious psychological patients for therapeutic practice. Starting from a A brief description of the case and/or scale scores returns a profile. comprehensive, realistic, and clinically consistent, incorporating nonverbal expression, Variability between sessions and principles of realism. Its output is structured in JSON format for direct import into PsychSim.
- **Psychopilot (debugging assistant):** prompt that configures the AI engine in transparent mode, directly addressing the therapist or developer without to embody any patient. Its function is diagnostic: it explains why a patient virtual responded in a certain way, identify which part of a system Prompt generates unexpected behaviors, suggests improvements, and reports them. limitations of the model (for example, failures in the generation of the scores on a scale).

4.4.2. Design of an automatic tester for the evaluation of LLM models

An automatic tester was also designed: a low-load test client that sends controlled requests to the PsychSim server to systematically evaluate the performance of different language models in the role of virtual patient, with the objective to identify those that would show the most consistent results with the tool requirements. The system integrates a test suite of clinical scenarios and edge cases (including scale management) and a self-session mode in which the The model itself acts simultaneously as patient and therapist, self-completing a self-characterization and a scaling process. This tool allows identification failures, inconsistencies, and operational limits of each model (limit testing), facilitating the validation of PsychSim as a research instrument.

4.4.3. Interviewed in Carol Ryff's Psychological Wellbeing Model

I participated as an interviewee in the adaptation and implementation process from Ryff's Model of Psychological Well-being (1989) to the assessment protocol of PsychLab. Specifically, I was administered the Psychological Well-being Interview (PBI, (adapted by Miriam Nieves), approximately four hours long, aimed at

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to explore how technological procrastination affected my health and overall well-being, focusing on the different dimensions of Ryff's model (positive social relationships, personal resources, autonomy, purpose, etc.). The objective was to assess the degree of impact of distress on overall well-being and integrate this model into the assessment from PsychLab. This experience "from the other side" of the research setting allowed to experience firsthand how the semi-structured interview works, to understand the participant's subjective experience in the Psycholab protocols and the tools that a good clinical interviewer should have in their repository.

Approximate distribution of the 225 hours

Period	Hours	Main activity
December 2025	~30 h	Conferences, PsychoCoffee, technical fine-tuning and start of MOOC
January 2026	~80 h	MOOC, Specialization Course and course feedback introduction
February 2026	~75 h	Self-assessment with PsychLab and project design procrastination
March 2026	~40 h	Project closure, Ryff interview, PsychSim testing and drafting of the report
TOTAL	**225 h**	

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5. Research Report

5.1. Introduction and theoretical framework

Adherence to behavioral change, and especially resistance to it, constitutes one
of the central problems of contemporary psychological practice. A good part of the
Interventions based on purely behavioral or motivational approaches are
They face high relapse rates, especially when the target behavior is
linked to profound aspects of the subject's self-concept. Faced with this scenario, the
The constructivist paradigm offers an explanatory framework that addresses change not as a
not merely a behavioral reorganization, but a reconfiguration of the system of
personal meanings that sustain the individual's identity.

Personal Construct Psychology (PCP), formulated by Kelly (1955/1991),
He understands the human being as a "scientist" who anticipates events through
The development of personal theories about their reality. Its Fundamental Postulate
It establishes that a person's psychological processes are channeled through the ways in which
This anticipates events, and they unfold through eleven corollaries among the
which highlight the Dichotomy Corollary—according to which the system is composed of
bipolar constructs—and the Choice Corollary, which posits that the subject chooses the
an alternative through which it anticipates a greater possibility of extension and definition of
his system. Within this cognitive architecture, Kelly distinguishes between constructs
peripherals—modifiable beliefs that do not alter the deep identity—and constructs
nuclear, which form the foundations of the self and govern the maintenance of coherence
internal aspects of the individual.

Within this framework, Hinkle (1965) expanded the model by introducing the concept of
implicative dilemma: a situation in which the desired change in a construct (movement
(from the symptomatic pole to the ideal pole) correlates, within the subject's idiosyncratic network,
with an unwanted movement in another nuclear construct. From this perspective, resistance
Change is not interpreted as a volitional deficit or a lack of motivation, but as
a logical system protection mechanism: the individual prefers to maintain the symptom
if its modification threatens core identity constructs.

The mathematical and computational approach to these structures has undergone a
Remarkable development in recent decades. The application of Network Science to the study
of psychopathology (Borsboom et al., 2021) and the incorporation of Fuzzy Logic

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through the Fuzzy Cognitive Maps proposed by Kosko (1986) have allowed
formalize the dynamics of the construct system in terms of weighted graphs,
susceptible to computational simulation. It is within this context that the
PsychLab virtual laboratory (Sanfeliciano and Saúl, 2022; Saúl, Botella and Sanfeliciano,
2023), which integrates classic idiographic techniques such as the RepGrid (Restruct Grid)
and the Weighted Implications Grid (WimpGrid) with advanced topological analyses (PB centrality
indices, density, PVAFF, Similarity to Ideal Index —SSI—) and
dynamic simulations of change.

This report articulates this approach in two complementary studies that
They operate at different but methodologically convergent scales. First, they
It presents an idiographic pilot study consisting of the self-assessment of one's own
researcher on the construct of academic self-demand and its contingency with the
stress (section 5.2). This study serves a dual purpose: on the one hand, it validates the
Viability and sensitivity of the PsychLab methodology for detecting and monitoring
real processes of change; on the other hand, it offers a proof of concept of clinical utility
of topological indices in case formulation and intervention design. In
Secondly, and based on the lessons learned from the pilot project, a project is presented
quasi-experimental study on technological procrastination, personality, and systems
personal constructs, designed as a team within the framework of the Practicum and oriented towards a
collective problem characterized by high relapse rates despite availability
of behavioral interventions (section 5.3). The integration of both studies allows
to demonstrate PsychLab's potential to bridge the gap between the idiographic and the
nomothetic.

5.2. Idiographic pilot study: academic self-demand and stress

Aim

The pilot study pursues a dual objective: on the one hand, to subject the methodology
PsychLab to a feasibility test under real-world conditions using a case study design
unique (N = 1); on the other hand, detect and formulate the dynamics of the construct system
personal factors associated with the experience of academic stress —understood from the model
transactional as an imbalance between perceived demands and resources
coping (Lazarus and Folkman, 1984; Barraza Macías, 2007)—, as well as evaluating the
effect of a brief intervention on said system.

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Method

Subject. Female, student of the Degree in Psychology (sociodemographic data) anonymized). The researcher acts as the sole subject, in accordance with the Practicum ethical guidelines that recommend the use of self-assessments or patients virtual to avoid formal clinical evaluation by students.

Variables and techniques. Two idiographic techniques based on the Theory were used of Personal Constructs:

- **RepGrid** for the elicitation and qualitative evaluation of core constructs linked to academic performance, self-esteem, and regulation emotional, with special attention to the dimension of self-demand and the self-oriented perfectionism (Hewitt and Flett, 1991).
- **WimpGrid** for mapping the network of implications between constructs and the identification of implicative dilemmas.

Topological analyses were performed in PsychLab (Sanfeliciano and Saúl, 2022; Saúl, Botella and Sanfeliciano, 2023): calculation of the PB indices —Presence and Balance of Implications—, Adjustment to the Ideal, Similarity to the Ideal Index (SSI) and Inconsistencies of the Ideally, complemented with RStudio for the representation of digraphs.

Procedure. The study was developed in three consecutive phases throughout approximately four weeks:

1. Pretest evaluation (baseline): writing the self-characterization, elicitation triadic constructs, RepGrid and WimpGrid administration, and analysis initial topological.
2. Case formulation and intervention design: identifying the dilemma central implication and development of an individualized narrative action plan ("Be like water"), along with a system of its own monitoring scales.
3. Post-test evaluation: re-administration of RepGrid and WimpGrid after two weeks implementation of the plan, recalculation of the topological indices and analysis pre/post comparison.

Results

System structure. The RepGrid yielded eight core constructs linked to the performance and self-demand: organized/disorganized, hardworking/lazy,

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efficient/inefficient, perfectionist/botched, relaxed/stressed, results of
10/normal results, tolerance for uncertainty/no tolerance for uncertainty and
Control/No Control. Centrality Analysis Using the PB Index (Appendix II)
It identified the construct "control" as the most influential node in the network: it presents the
a greater number of outputs and a strongly superordinate balance, which suggests its
nuclear nature and its high resistance to change.

Central implicative dilemma. The WimpGrid analysis revealed a critical implicative dilemma.
(see the system digraph, Appendix II): the reduction of the "control" pole—that the subject
You wish to modify it to achieve greater relaxation—this would involve shifting it towards the non-polar pole.
desired of "disorganized", identity-congruent construct (actual-self = self-
(ideal). This configuration turns stress into a functional "price" paid by
Maintaining identity coherence: discomfort does not operate as a dysfunctional effect
residual, but as an active mechanism for protecting the construct of "organized"
in the face of the perceived threat that relaxing controls would pose. This finding
This corresponds to an inconsistency of the Ideal detectable in the digraph of the Ideal-Ego, in the
that achieving one congruent goal would simultaneously mobilize the opposite pole of another
equally valued objective.

Pre/post comparison. After two weeks of implementing the intervention plan, the
Indices show a relevant structural advance (Appendix II):

- The Similarity to Ideal Index (SSI) went from 0.38 (low fit situation and
  high self-demand) to 1.00 (full adjustment to the Ideal Self), as reflected in the map
  of heat from Figure 3. The pretest value suggested an initial moment of
  system reorganization or an excessive level of aspiration; the post-test indicates
  an effective reconfiguration.
- The Ideal Fit graph (Figure 2) shows a generalized shift
  from constructs towards the ideal pole. Some reach it or surpass it.
  (hardworking, organized, tolerant of uncertainty); others, such as
  Whether perfectionist or efficient, they already started from high levels and show stability
  with an upward trend. The constructs that started from the negative pole (results of
  10) or very close to neutral (no control) advance to values close to 0.5.
- The Impact-Feedback analysis (Figures 4 and 5) shows a redistribution of the
  constructs between pretest and posttest: in the posttest, the constructs associated with

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performance reduces its disproportionate impact on the system, which suggests greater coherence and greater internal balance after the intervention.

It should be noted that some constructs (perfectionist, results of 10) present values extreme ideals, which suggests reviewing whether they respond to realistic goals or reflect a persistence of self-demand which, even having lost centrality, retains weight aspirational.

Discussion

The pilot study fulfills its dual objective. First, it validates the sensitivity and the Clinical utility of the PsychLab methodology: topological indices detect both the core structure of the problem (centrality of control, implicative dilemma) as the effective change after the intervention, offering a mathematical and visual representation of the process. Secondly, it reveals the functional nature of the discomfort within the system, in line with the conceptualization of self-oriented perfectionism as a a trait that, despite its high emotional cost, supports goals of self-esteem and competition (Hewitt and Flett, 1991). This has direct implications for the design of interventions: working on the symptom without attending to the identity constructs that argues that it would be counterproductive or, at best, ineffective in the medium term.

These two conclusions—methodological feasibility and the importance of mapping the network—identity that sustains the target behavior— constitute the foundation of the second study that is presented below, in which the same approach is applied to a phenomenon collective characterized by high resistance to change: technological procrastination.

5.3. Quasi-experimental project: technological procrastination, personality and system of personal constructs

Specific justification

Technological procrastination—understood as the maladaptive and purposeful use of technology—is a tool for learning and learning. Avoidance of digital devices—constitutes an emerging phenomenon whose approach From purely behavioral or motivational approaches, it presents an effectiveness limited. The traditional approach, anchored in the notion of a self-regulation deficit or willpower, fails to explain the high relapse rates observed after Conventional psychoeducational interventions on digital hygiene. The literature Contemporary approaches suggest distinguishing between the quantity of use (screen time) and the quality

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of the same (active versus passive use), as well as considering the role of Fear of Missing Out (FoMO) and digital emotional regulation processes as strategies dysfunctional avoidance patterns (Villalvir Santos and Costa Ortega, 2024).

The present project is justified by its methodological effort to integrate the evaluation nomothetic of universal features with idiographic mapping of personal meanings. This dual lens is necessary because technological procrastination is unsustainable. not only on a vulnerable biological basis, but its maintenance can In many cases, this can be explained by their protective function within the system of constructs. from the subject's perspective: abandoning the habit would threaten core constructs such as social connection, belonging or relational availability, which activates resistance to change characteristic of implicative dilemmas (Hinkle, 1965; Senra et al., 2005).

Specific theoretical framework

The theoretical approach is structured around two complementary pillars, placing the Theory of Personal Constructs as the structuring framework of the study and the model psychobiological as a complementary nomothetic contribution.

(1) Personal Construct Theory (Kelly, 1955/1991; Botella and Feixas, 2008). As a main framework and in accordance with what is developed in section 5.1, the PCP maintains that behavior does not change in a stable way without altering the network of meanings that the anchor. Applied to the digital realm, procrastination would persist not because the subject not because they lack motivation, but because their cognitive system anticipates that the disconnection would invalidate core constructs of the self (being "connected" ÿ being "integrated", "informed", "relevant"). Mapping these constructs using the WimpGrid and the Topological analysis in PsychLab allows the identification a priori of implicative dilemmas that predict resistance to behavioral adherence. From this perspective, resistance to Digital disconnection does not constitute a volitional deficit, but a protective mechanism of identity.

(2) Cloninger's psychobiological model (Cloninger et al., 1993; standardized in population Spanish by Gutiérrez-Zotes et al., 2004). As a nomothetic complement, this model distinguishes between neurobiologically based temperament dimensions (Search for Novelty, Harm Avoidance, Reward Dependence, Persistence) and dimensions of character modulated by learning (Self-direction, Cooperation, Self-transcendence). In the digital context, a high Novelty Seeking phenotype

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Combined with low persistence and self-direction, this would create a vulnerability profile:
The exploratory need predisposes one to immediate gratification from the digital environment,
while the deficit in character resources makes it difficult to inhibit impulses in the face of
Persuasive designs of contemporary applications. The integration of this lens
Nomothetic methods combined with Kelly's idiographic mapping allow us to transcend generic approaches
focused exclusively on behavior.

Objectives and hypotheses

General objective. To evaluate the influence of temperament and identity structure
in modifying technological procrastination, analyzing the behavioral impact and
cognitive effects of a 14-day digital hygiene intervention in regular adult users
smartphone.

Specific objectives:

4. Characterize the psychobiological profile of the participants in the pretest phase
   using the TCI-R-67, determining its Novelty Search levels,
   Persistence and Self-Direction.
5. Elicit and model the identity structure of the subjects through
   Self-characterization and WimpGrid, digitizing its construct system
   personal in PsychLab.
6. Record behavioral and emotional metrics (target screen use,
   FoMO (fear of missing time) before and after the intervention.
7. Compare the pre/post change in the topological indices yielded by PsychLab
   (density, PB centrality, inconsistencies) and determine if the cognitive structure
   The initial one acts as a predictor of resistance to change.

Hypothesis:

- **H1 (nomothetic and cross-sectional):** participants with a high Search profile
  Novelty combined with low Persistence and Self-Direction will yield higher
  Baseline levels of FoMO, technological procrastination, and emotional distress in the
  pretest phase.

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- **H2 (longitudinal):** digital hygiene intervention will produce a reduction significant in objective metrics of smartphone use and in dissatisfaction perceived, accompanied by an observable topological restructuring in the system of personal constructs.
- **H3 (predictive idiographic):** the structural variables of the PsychLab in pretest —specifically the high density of the system and the presence of dilemmas central implicants—will act as statistical predictors significant factors in the low adherence to the intervention and the smaller reduction in levels of technological procrastination.

Method

Design. Quasi-experimental, longitudinal, repeated measures (pre/post), with two groups: an experimental group (which will receive the digital hygiene intervention) and a group Baseline (active control). Group assignment will be made after the evaluation. pretest based on initial technology interference in life scores daily.

Participants. Non-probability convenience sampling, through recruitment via social media, university settings, and psychological care centers. Criteria for Inclusion: legal age, sufficient verbal fluency to conduct an interview Deep semi-structured survey and regular smartphone use. The final sample size It will be determined by an a priori power analysis with G*Power (Faul et al., 2009), assuming standard parameters for a repeated measures ANOVA; to the calculation The resulting amount will be increased by a safety margin to compensate for mortality. experimental characteristic of 14-day longitudinal interventions.

Variables. The independent variable is group membership (experimental versus (baseline). The dependent variables comprise the features of the Cloninger model. (TCI-R-67), the topological indices of the construct system (density, PB centrality, inconsistencies), target screen time and FoMO scores, Academic procrastination and self-reported emotional distress were identified as covariates. They record age, gender, and academic or employment status.

Tools:

- **TCI-R-67** (Pedrero Pérez, 2009): abbreviated and validated version of the Inventory of Cloninger Revised Temperament and Character Scale, 5-point Likert scale, with

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solid internal consistency (Cronbach's alphas between 0.80 and 0.83 in the dimensions clue).

- **Academic Procrastination Scale (EPA)** in its version adapted to Spanish (Trujillo-Chumán and Noé-Grijalva, 2020).
- **Self-characterization** according to Kelly's original framework: short story in the third person person from the perspective of a close and understanding friend, designed to minimize self-criticism and elicit the poles of similarity and contrast.
- **WimpGrid:** Semi-structured interview for network elicitation and evaluation weighted implications between personal constructs.
- **PsychLab software:** topological analysis and dynamic simulation of the system constructs.
- **Objective screen usage metrics** extracted from the operating systems of the terminals (minutes of daily use, number of unlocks), as a contrast objective to self-reports.

Procedure. It will be developed in three consecutive phases. In Phase 1 (pretest), after the Signing of informed consent, participants will complete the TCI-R-67 and the EPA, They will draft the Self-Characterization (from which the idiosyncratic constructs will be extracted), They will be interviewed using the WimpGrid and their system will be digitized at PsychLab; Finally, initial screen usage metrics and discomfort scales will be taken.
In Phase 2 (intervention, 14 days), the experimental group will follow a protocol behavioral digital hygiene articulated in four guidelines, each based on Specific mechanisms of self-regulation and neurobiology:

8. **Disabling non-essential notifications:** notifications operate as exogenous stimuli that hijack attention under a reinforcement schedule intermittent with variable intervals, consolidating the habit of compulsive checking; Deactivating it returns attentional control to the user.
9. **Closed usage windows** (2-3 daily accesses with time limits): technique of Stimulus control based on time-boxing, which reduces partial attention Continue and train your tolerance for delayed gratification.
10. **Intentional pause:** explicit verbalization of the question "Why am I entering?" before unlocking, which activates top-down executive control and introduces friction cognitive in automatic loops.

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11. **Circadian attentional hygiene:** 30-60 minutes without screens upon waking and before of sleep, which protects melatonin secretion and lessens the impact immediate after the morning FoMO.

The baseline group will maintain its usual habits with passive metric tracking. Phase 3 (post-test) will involve re-administering the WimpGrid, the screen metrics log. and self-report scales, to quantify both behavioral modification and possible topological restructuring of the system.

Analysis plan. Descriptive analysis of sociodemographic variables and of the TCI-R-67 scores; Pearson correlations for H1; mixed ANOVA of repeated measures (group × time) for H2; and multiple linear regression models, with pretest topological indices as predictors of the magnitude of posttest change, for the H3.

Expected results and discussion

At the nomothetic level, the correlation between Novelty Seeking and increased problematic use of screens, as well as the inverse relationship with Autodirection and Persistence, supporting the clinical utility of Cloninger's model in the field of behavioral addictions.

At a longitudinal level, a significant reduction in objective metrics is anticipated. smartphone use and self-reported discomfort in the experimental group, accompanied of an observable topological restructuring in the system of constructs and of a Increased insight compared to previous use. Confirmation of this result would support the constructivist premise that all sustained behavioral change requires, inexorably, an alteration in the way the subject anticipates and constructs his reality.

At an idiographic-predictive level, confirmation of H3 would give empirical weight to TCP in predicting therapeutic success: if the pretest mathematical indices (high density, implicative dilemmas) predict resistance to change, it would be evident that the difficulty The act of "disconnecting" is not a response to a volitional deficit, but rather to an active mechanism of protection of identity. Such a finding would open the door to psychological interventions. individualized, where the goal is not simply to reduce screen time, but rather to resolve the underlying identity conflicts that maintain the target behavior.

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The main limitation of the proposed design lies in its nature as a project: the present
The report describes the complete protocol, but data collection was not included in the
Practicum timeline. This limitation can be remedied in the immediate future, since the
The protocol has been fully operationalized (consents, scales, templates).
WimpGrid, interview script and scoring spreadsheets; see Appendix III).

Ethical considerations

The research was designed in accordance with the ethical principles established by the committees.
university students and international guidelines on research with human subjects.
It includes an informed consent form that guarantees the right to leave the study.
at any time without penalty; anonymity and confidentiality through
pseudonymization of identifying data with unique alphanumeric codes; the
principle of beneficence, with an exit interview after the post-test to ensure the
emotional well-being of the participants and their referral in case of detected distress
clinically significant; and the processing of data in accordance with the GDPR, with UNED as
responsible and Dr. Luis Ángel Saúl as institutional representative.

5.4. Integrative Conclusion

The two studies presented in this Report, far from being independent works,
They form a single storyline that runs through the idiographic and nomothetic scales.
The pilot study (5.2) demonstrated, on a single case and with real data on change, that
The PsychLab methodology is capable of detecting the core architecture of a problem —
the centrality of control, the implicative dilemma between "control" and "organized"—and of
mathematically monitor an effective change process (SSI from 0.38 to 1.00 after a
(brief intervention). The most relevant aspect of the pilot was not the improvement itself, but the evidence
that malaise served a protective function within the system: a conclusion that
It completely reorients the design of any intervention.

This same idea forms the backbone of the project on technological procrastination (5.3). If
In the self-case, stress operated as a "price" to sustain an organized identity,
The project's central hypothesis argues that digital procrastination persists in
many subjects, because disconnection would threaten core constructs linked to the
belonging and social relevance. In both cases, the symptom is not a system failure,
but a functional part of it.

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Thus, the report's trajectory illustrates PsychLab's potential to develop a
bridge between the individual and the collective: what is validated on the system itself
Personal meanings can be projected, with the same methodological rigor, onto the study
of collective phenomena resistant to change. The overarching conclusion is clear: everything
Sustained behavioral change requires a reorganization of the network of meanings that it
It maintains that only interventions capable of operating at that level achieve results that
They endure.

6. Personal assessment

The completion of this Research Practicum has meant, beyond mere compliance
academic, a profoundly transformative experience that has allowed me to tend
a real bridge between the theoretical framework studied during the Degree and scientific practice and
researcher.

The aspect I consider most unique about this experience is having gone through the process
from both sides of the research setting. On the one hand, as a researcher, I learned to
operationalize concepts, design protocols and manage the technical tools of
laboratory (Jamovi, RStudio, Markdown and GitHub), as well as to apply and interpret the
idiographic techniques (RepGrid, WimpGrid) and PsychLab's topological indices. By
On the other hand, when I carried out my own self-assessment, I experienced firsthand what
It means confronting one's own system of constructs: understanding that my discomfort was not
not a simple "flaw" to correct, but a functional piece that protected my identity, changed
completely changed my understanding of resistance to change.

Seeing the implicative dilemma represented in the digraph was revealing in a way that
Introspection alone hadn't given me the insight to understand that those two constructs were so
nuclear and so interdependent —that moving one directly threatened the other—
It gave a name and form to something that until then had only existed as bodily tension. Something
Something similar happened with the SSI: the initial 0.38 and the final 1.00 weren't just numbers, they were a
a mathematically accurate representation of how I felt before and after. That a
The fact that the index captured something so subjective with such fidelity was probably the most crucial test.
convincing evidence of the methodology's validity. Methodologically, the design in
The procrastination project team has consolidated my research maturity.
I overcame the challenge of formulating a complex quasi-experimental project, integrating variables
biological (Cloninger model) with the phenomenological and mathematical analysis of

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system of constructs. The experience of working in coordination with the rest of the students, distributing tasks and building a coherent protocol among several people has been especially valuable.

The testing phase of PsychSim also forced me to develop a critical perspective towards The use of Generative Artificial Intelligence in psychology. When creating the prompts of By studying virtual patients and designing the automated tester, I understood firsthand that these Models, however sophisticated they may seem, operate by statistical prediction and lack Real understanding: clinical interpretation and the design of "bridges of change" Therapeutic approaches always depend on professional judgment and the constructivist theoretical framework. I learned that technology is an extraordinary simulator for training skills, but never a substitute for human sensitivity.

Also, participate as an interviewee in the adaptation of the Wellbeing Model Ryff's Psychological (1989) —which articulates well-being around six dimensions: self-acceptance, positive relationships, autonomy, mastery of the environment, purpose in life and personal growth—it allowed me to appreciate the importance of nurturing the experience of participant in any evaluation protocol, a perspective that is hardly acquired from the exclusive role of researcher.

Overall, this Practicum has significantly expanded my vision of psychology. based on evidence and has confirmed my interest in constructivist research as a way to address human suffering rigorously and, at the same time, with respect for the uniqueness of each person.

This experience has further defined a professional direction that I already suspected but that The Practicum has made concrete: the intersection between psychology and technology. Working with mathematical indices on the construct system, designing prompts for language models and explore how deep learning can quantify meanings Personal experiences have confirmed for me that formal rigor is not at odds with understanding. subject—on the contrary, it empowers it. I take both the constructivist methodology and the technical skills and, above all, the vision that both can and should engage in dialogue. For the final degree project, the possibility of integrating objective measures —neurosiological, mathematical, computational—with the idiographic analysis of subjective experience is a line that It deeply appeals to me, and this practicum has given me concrete tools to explore.

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8. Appendices

The appendices referenced throughout this document are listed below. Inclusion is justified by its relevance to understanding the work done and the results reported in the Research Report.

Appendix	Content
Appendix I	Review and feedback report of the MOOC "Introduction to PsychLab".
Appendix II	Complete idiographic pilot study: self-characterization, formulation of case, therapeutic return and figures of topological analysis.
Appendix III	Complete research project on procrastination, personality and personal constructs.
Appendix IV	PsychSim Prompts: "Dr. JSON" and "Psychopilot".
Appendix V	Design of the automatic tester for the evaluation of LLM models.

Document organization criteria:

To optimize the structure of the Appendices and avoid unnecessary repetition of documentation, those materials that are already incorporated in Appendices of Related projects or activities are not presented again as Appendices independent, maintaining only the reference to their corresponding location.

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APPENDIX I. Review Report



MOOC REVIEW REPORT

“Introduction to the use of PsychLab (3rd ed.)”

Lucía Dovale Posada

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I. Introduction

This report will record my experiences and feelings as a student taking the course “Introduction to the use of PsychLab (3rd ed. 2025-2026)” for the first time, focusing mainly on the learning process and the understanding of the explanations with the
The aim is to contribute their bit to possible areas for improvement.

Regarding my student profile, it's worth noting that I've already completed another Java course on the edX platform, so the interface is familiar to me from the start. Furthermore, as a Psychology student at UNED, I'm already familiar with handling statistical data and interpreting graphs, as well as other psychological and mathematical concepts. I'd like to highlight this information while also
I question the ability, or rather the ease, of a user less familiar with this situation to navigate the course in the same way.

II. Summary

Overall, the MOOC is intuitive and well-structured, especially for a first introduction to PsychLab. I found it very helpful, particularly because of the way it brings together the various learning materials available.
Psychlab and the Grid Technique, while extremely useful, can make tracking progress confusing due to the fragmentation of the material, leaving one unsure of how to track progress. I'm very good at knowing where you stand and what your knowledge is, but that's not the case. This way, the route is well guided. On the other hand, I believe that some points might raise questions for some users, as I will discuss later.

III. Clarity and understanding

As a first-year student, what perhaps caused me the most effort was not so much the explanations or procedures, but rather the assimilation of all the new concepts, lacking access to, for example, a glossary or a *tooltip* (a pop-up message with a definition or link to a previous section) for certain concepts. Both methods would significantly reduce the user's cognitive load since, not only would they not
They wouldn't slow down the learning flow or increase fatigue, but would be there for those users who needed them, improving progress and reducing minor issues.
interruptions to go back and manually search for the information.

I found most of the definitions and practical instructions clear and precise. On the other hand, I would focus particularly on the
the “mathematical” part, such as: reading graphs or charts, statistical concepts…dealing
I approached the situation of a user with limited training, since in my experience that would be the only thing that is somewhat "taken for granted" and is usually the least intuitive. I didn't have too many problems following the application protocol.

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from the grid (Unit 3: Tasks - Practical Exercise), however, something that I did notice The part that significantly hampered the evaluation was the bias identification aspect, since As a beginner, I couldn't properly extrapolate the different nodes and their causes. consequences. Something that would have helped me would have been access to a wider variety of grids or an interview, as well as some illustrative protocol of the errors. common or “what not to do”.

From a scientific perspective, it seems to me that the choice of study methodology The advanced Psychlab course is particularly beneficial. Among the most widely used and supported techniques, spaced practice stands out ; that is, studying the same content in several separate sessions. This improves long-term learning in many subjects and age groups (Dunlosky et al., 2013).
This method is reflected in the mandatory pause caused by having to wait for the review of assignments, in addition to the fact that the content is repeated and progresses. constantly as the course progresses. Furthermore, the use of Active Practice *(practice testing)* is also noteworthy . Asking questions, taking tests, or reflecting without looking at notes significantly improves performance at different educational levels and in various subjects (Dunlosky et al., 2013). This is reflected in the course when students have to reflect on the content, for example, when raising a discussion topic in the forum, or when they have to extrapolate the theory to apply it to clinical practice or to graphs. Meta-analyses confirm that Active Practice is one of the techniques with the greatest impact on learning (Donoghue & Hattie, 2021). Finally, Interleaved *Practice,* understood as mixing different types of problems instead of always doing the same one, also promotes learning (Dunlosky et al., 2013), highlighting its importance.
having to analyze different clinical cases and graphs throughout the course.

IV. Usability and experience

Regarding navigation through the course, I felt well-placed at all times, aware of where I was and what my next objective should be. However, in my opinion, the training would be improved if the supplementary materials were located "within" the course itself, that is, in the relevant sections or where the most questions might arise. For example, in the more challenging mathematical sections (Unit 2: Studying Psychological Change with Psychlab - Mathematical Modeling), the interactive manual "Mathematical Modeling" could be used, as well as in sections that might present a challenge. For instance, in this latter case, I believe the interactive resource "Interpreting Results: Implications Grid" would be extremely valuable in the sections dealing with statistical results.

I believe that by placing all these resources in a section near the end (Supplementary Material), some potentially useful resources might be overlooked until the end of the course when they are no longer needed. While the material is certainly accessible at all times, I generally believe that the progression of these types of courses encourages a linear, end-to-end approach. Therefore, I think that some of the material, despite its significant educational value,

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This would be reduced by its position.

This is a strategic placement of the teaching resources that we can observe as the course progresses, and I think it greatly enhances the progress and navigation, especially at the beginning, when you are most lost.

V. Formal aspects

Some minor errors were detected, as well as some specific doubts, which do not seriously affect the general understanding, but which could be corrected to improve the presentation and flow of the material.

- For the typo "...influyen y quina más abajo, más indirectamente…", see Unit 4: Analysis of a WimpGrid - Analysis of digraphs, last line of the paragraph of "Multidimensional Scale (mds)".

- Regarding a question, in question #3 of Unit 4:

  Or, if we want to know which is the most influential construct in the system, which index should we consult?

  ÿ The degree and number of inputs.
  ÿ **The degree and number of outputs.**
  ÿ Proximity.

  From my point of view, I believe that this question gives rise to doubts because I initially considered both A and **B** ("The degree and number of inputs" and **"The degree and number of outputs") to be correct.**

  Given that the **degree** "is defined as the number of edges (arrows) that are joined to each of the vertices (nodes or constructs)", and "can be estimated from the amount of *inputs* and *outputs* in each of the constructs" (see Centrality Analysis) and the **influence** of a node is its ability to affect other nodes and is obtained by adding the number of outputs.

  Therefore, my approach was as follows:

  To calculate the most influential construct, I need to know which one has the highest number of *outputs.* For this, I must be familiar with this index (form 1), the *input index ,* and the degree (form 2):

  ÿ Form 1: Knowing the amount of *outputs* directly would already indicate the influence, I would just have to compare it with the other nodes.
  ÿ Form 2: I would apply the following formula with which I could then proceed according to the form 1

  degree - *inputs* = *outputs*

  Therefore, following this method, both A and B could be true. Even,

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> Option A, "The Degree and Number of Inputs," could be considered "more accurate," since in this case we do need both elements to determine influence, whereas in option **B , "The Degree and Number of Outputs,"** we would only need the number of *outputs,* according to the information provided: "those with a higher number of outputs are the most **influential** in the system." If we were talking about *balance, presence, etc.* (see section "Centrality Analysis"), I could understand that we would need the degree to determine the output/input ratio or some other criterion. I understand that perhaps what leads to To consider B as true is to assume that the most influential nodes are also considered "nuclear" in and of themselves, that is, those with a greater number of outputs and a lower number of inputs, These being the most influential and the least influenced. However, my experience as a student in the course was different. These doubts could benefit from having feedback on the correct answer when correcting self-assessments, as happens in the advanced course.

Finally, in question 4, “What might an inverse relationship between two poles of the ideal be indicating?” I would find the use of the term “nodes” more natural than “poles,” since the latter leads me to think of the poles of the same construct and not the “poles” that define the ideal, since the “nodes of the ideal” would already be understood.

VI. Future proposals

As potential improvements, it might be helpful to add methods to reduce the cognitive load of the course, especially for first-time students, such as the aforementioned glossary or *tooltips.* An alternative location for supplementary materials could also be suggested, and perhaps some sections of text could be partially replaced with more visual, interactive material. This material would contain the same information as the text but in a more engaging and playful way, as is typical of these interactive presentations. For example, one of the existing manuals in the "Supplementary Material" section could be used, supplementing it with any additional information not already included.

Another consideration that might be helpful for some users would be to include a short explanatory justification in the self-assessment questions, or a reference to the section to be reviewed in the failed question to promote active feedback.

VII. Conclusion

In essence, and after completing the next Psychlab course, positive changes that could be considered would include, for example, those already implemented in the next course (glossary, instruction manuals, etc.). Furthermore, we anticipate a positive reception from students, given that they are currently functioning correctly in the advanced course.

Overall, the MOOC offers a solid introduction to PsychLab and is a good starting point for students with no prior experience. The feedback gathered in this report aims to provide constructive feedback from a first-time user's perspective.

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APPENDIX II. Pilot ideographic study

I. Critical Review: Advantages of the Network Approach in Psychology

In recent years, traditional explanatory models used to understand psychological distress have begun to prove inefficient in meeting the full demands of current clinical psychology, showing themselves to be obsolete in certain areas.
and sparking criticism from the psychological community. In this context, the Network Approach emerges as a theoretical and applied alternative that proposes a different way of addressing psychological processes. This text will compare the traditional categorical model, represented by manuals such as the DSM, with the Network Approach.
emphasizing why the latter is more useful when it comes to capturing and understanding the complexity and dynamism of psychological processes.

The categorical model, represented by the DSM (Diagnostic and Statistical Manual of Mental Disorders), organizes mental disorders around diagnostic categories defined by relatively invariable specific criteria. It is based on medical logic, characterized by treating symptoms not as the cause, but as observable manifestations of an underlying disorder. Although this system has initially proven useful in defining different mental disorders and providing
Standardized diagnoses and organizing the complexity of mental health into structured categories, as well as facilitating communication between professionals by establishing a common and interdisciplinary language, have received significant criticism.

Among the limitations of this model is the high comorbidity between
diagnoses, derived from unsuccessful attempts to create a detailed and precise system, which, instead, has resulted in a strong overlap of repetitive diagnostic criteria and the arbitrary separation into rigid categories of entities that may be different manifestations of a single underlying disease; it is also worth highlighting the poor suitability to adapt to a continuous clinical nature, not being able to justify how psychological problems develop and are maintained over time.
Thus, the categorical model tends to simplify complex and dynamic processes, reducing them to relatively static, simple diagnostic labels.

In contrast to the traditional model, the Network Approach proposes an alternative way to understand psychological distress, drawing inspiration from Graph Theory. From this model, psychological problems are described based on the interaction between different elements. Elements, called nodes, can be symptoms, thoughts, emotions, or behaviors. The relationships between these nodes, called edges, reflect how the nodes influence each other, articulating the dynamic, context-dependent system of personal meanings used to conceptualize the self and distress. Furthermore, these patterns will vary among patients, resulting from different personal experiences; they will not be uniform and will adapt to their individual characteristics.

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One of the main advantages of the Network Approach compared to the categorical model is the graphic representation of the patient's system of meanings and the structure of their distress, resulting in a closer approximation to personal experience and offering tools with
a clear therapeutic applicability, such as the dynamic maps of Psychlab. This allows for optimal analysis of psychological change and serves as valuable material for the professional.
of high functional value, offering, among other things, a clarifying visual guide and mathematical data to reveal underlying patterns or effects in relationships that might be overlooked at first glance. Furthermore, by focusing on the relationships between nodes, it is possible to observe how modifying a specific element can produce effects throughout the system and how the system functions as a whole. This facilitates the

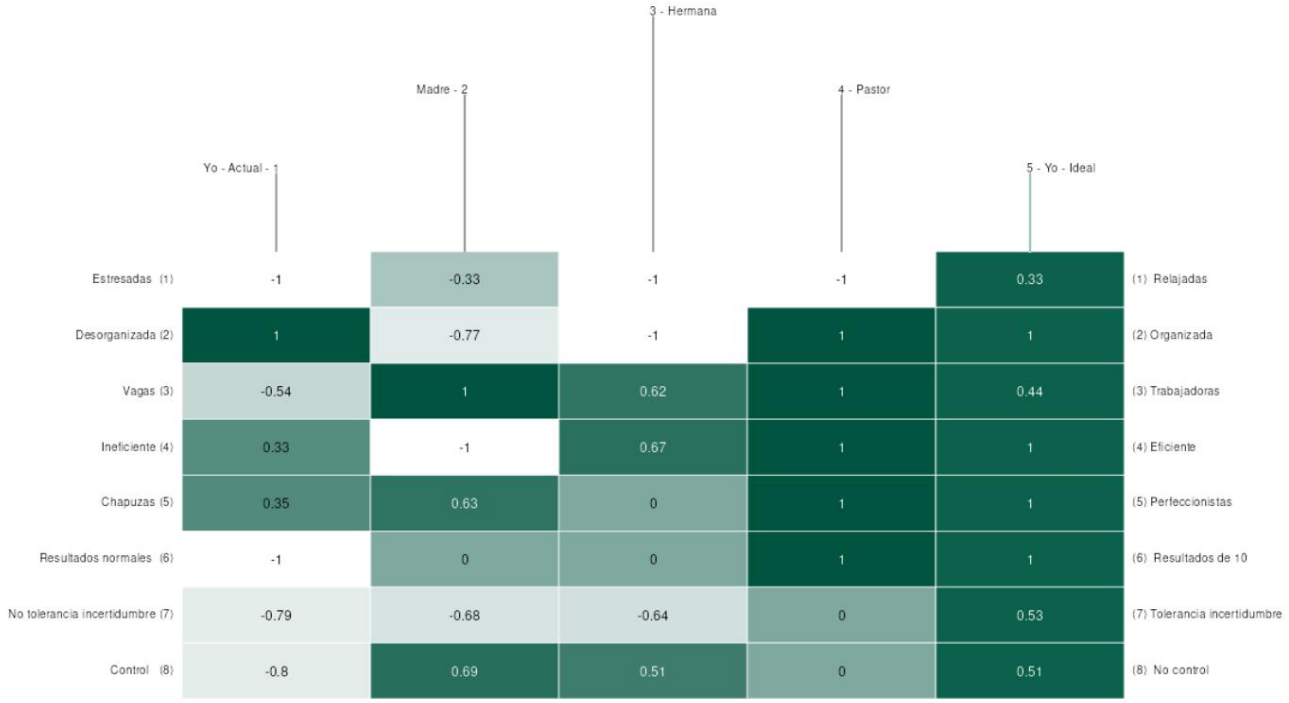
Personalized identification of relevant points, such as central nodes or rigid cycles, allows for a tailored intervention adapted to the patient's individual experience. Overcoming diagnostic rigidity enables the most precise and up-to-date monitoring possible at each stage of the patient's progress. Ultimately, both the professional's performance and the feedback the patient receives, which allows them to continue progressing, will benefit from the application of this approach.

In conclusion, the Network Approach offers significant advantages and advancements, with characteristics more aligned with current clinical psychology, providing a more dynamic, personalized, and holistic view of psychological processes. It stands as an especially valuable tool for clinical understanding and intervention, and also offers the possibility of being combined with other approaches to provide a multidisciplinary perspective. Its development encourages further exploration of new ways to represent and work with the complexity of human distress.

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II. Evaluation of Personal Meanings with the Grid Technique

I'm presenting some ideas for changes in my daily life that I'd like to implement, and to do so, I'll use my own personal case as the subject of the Grid. I'm a 21-year-old university student. I've been dealing with this stress for years, and I'd like to change the way I manage the stress I feel in situations like exams and those that make me doubt my own self-efficacy. After failing to manage this temporary stress, I've noticed that even after the stressful situation passes, it transforms into chronic anxiety that ends up overwhelming my coping strategies, which I previously used to handle effectively. To manage these situations, my coping mechanisms are usually based on worry, both conscious and unconscious, which creates limitations in my daily life. Despite trying different methods and feeling more psychologically relaxed, the presence of somatic symptoms persists across various situations.



To address this issue and create my Grid, I opted for selecting elements by choosing representative and important figures in my life. These include elements such as 'current self', 'ideal self', 'mother', 'sister', and 'shepherd' to represent their own experiential perspectives on coping with stress. My 'current self' represents a state of alertness and stress dependent on perceived performance, while my 'ideal self' represents a perspective in which, through a system of secure self-efficacy and greater fluidity, I achieve my desired goals—that is, less anxiety and better results. My 'mother' and 'sister' represent figures close to me who are also compatible, to a certain extent, in their methods of coping with critical situations, exhibiting both effectiveness and a degree of stress, as we will see later.

'Pastor' is a good friend who is also a student and experiences stress, but in a way that does not prevent him from achieving good results, reflecting a correct management of stress.

Regarding the selected constructs, a brief explanatory analysis is offered,

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highlighting the poles of the ideal:

'Stressed/Relaxed'	It represents the poles of stress vs. relaxation, central to the proposed problem, and which would include the somatic symptoms produced by anxiety.
'Control/no **control'**	State and feeling of negative alertness based on 'being controlling' ('control') or, on the contrary, being able to let things flow ('no control').
'Disorganized/organized'	Ability to plan time and achieve all proposed goals within a period of time.
'Lazy/working'	To have or to gain motivation to work.
'Inefficient/efficient'	Awareness of one's own abilities and their correct use
'Botched/perfectionist'	High standards and meticulousness in our work.
'No tolerance for uncertainty' / **'tolerance for uncertainty'**	Ability to flow in new or unknown situations.
'Results **normal/10' results**	Achieving good but improvable results, 'normal results' or, the goal, '10/10 results'.

The selected constructs can be grouped according to the following classification:

- Emotional themes:
    - 'stressed/relaxed'
    - 'control/no control'
    - 'lazy/hardworking'
- Identity issues:
    - 'disorganized/organized'
    - 'inefficient/efficient'
    - 'botch worker/perfectionist'
    - 'normal results/results of 10'
    - 'no tolerance for uncertainty/tolerance for uncertainty'

Based on the proposed grouping, the importance of identity-related themes as constructs of identity stands out, representing how the *self* is represented in
around constructs that will define the system of perceived self-efficacy and performance; leaving the emotional ones in the background, despite presenting a core construct for discomfort, 'stressed/ relaxed', predicting possible behavior as consequences rather than causes.

Despite the potentially ambiguous initial appearance and a certain repetitive profile, the constructs

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'Control/no control' and 'no tolerance for uncertainty/tolerance of uncertainty' would operate as characterizers of the capacity for flow. 'Control/no control' would represent the emotional aspect, such as the feeling of needing to be in control, and 'no tolerance for uncertainty/tolerance of uncertainty' the identity aspect, such as the self-perception of being able to flow. The construct 'stressed/relaxed' would show a certain rigid tendency towards
mainly focus on the negative pole with several scores on the pole
Negative. On the other hand, the 'disorganized/organized' construct is notably dichotomous, varying mainly between scores of 1 and -1. Finally, 'normal results/10 results', despite showing a markedly dichotomous aspect similar to the previous construct, also exhibits prevalence at the neutral point, 0, representing an unusual characteristic. Despite these characteristics, the BIAS index presents a score of 0.41, not indicative of a profile with polarized scores and cognitive rigidity, although localized rigidity in the aforementioned constructs cannot be ruled out.

Yo/Ideal	Yo/Otros	Otros/Ideal	PVAFF	Intensidad	Conflicts	BIAS
-0.148	0.002	0.286	0.466	0.292	0.679	0.41

According to the indices specific to the construction of the self ('self/ideal', 'self/others' and 'others/ideal'), representative of the information coming from oneself and from the
When we interact with others, we can speculate on several factors. The first index is 'self/ideal'
This would represent low self-esteem given its low value (-0.148), indicating the magnitude of the
The necessary change to reach the ideal could also be indicative of a high standard set by that ideal. The 'self/others' dimension would reflect a high level of self-perceived social isolation, given
the low perceived similarity with the others (0.002). The 'others/ideal' would show a certain
Perceived adequacy in others (0.286). Visualizing these indices together, we would be facing a tentative profile of isolation or resentment.

The network will be structured around the presence of the constructs 'organized', 'efficient' and congruent: 'perfectionist', although the latter two are not as strongly found

rooted in identity with
the 'organized' dimension and the discrepant ones: 'stressed', 'lazy', 'normal results', 'no tolerance for uncertainty' and 'no control'. to

Construct	Self	Ideal	Difference	Classification
Estresadas - Relajadas	-1.00	0.33	1.33	discrepant
Organizada - Desorganizada	-1.00	-1.00	0.00	congruent
Vagas - Trabajadoras	-0.54	0.44	0.98	neither
Eficiente - Ineficiente	-0.33	-1.00	0.67	neither
Perfeccionistas - Chapuzas	-0.35	-1.00	0.65	neither
Resultados normales - Resultados de 10	-1.00	1.00	2.00	discrepant
No tolerancia incertidumbre - Tolerancia incertidumbre	-0.79	0.53	1.32	discrepant
Control - No control	-0.80	0.51	1.31	discrepant

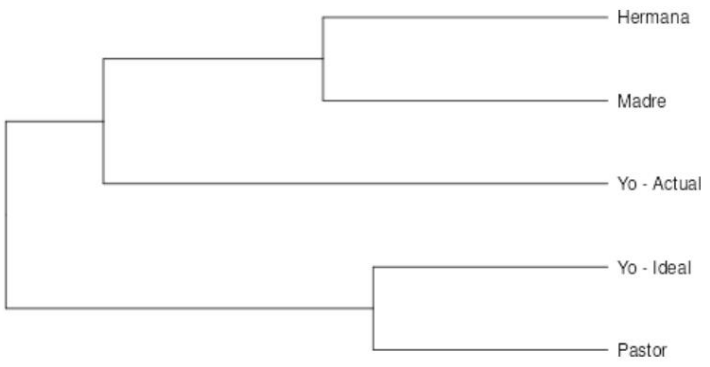
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These latter factors reflect the cycle of discomfort, representing the weak points of identity and the dimensions that one wishes to change. Therefore, stress would not be the primary problem. but a protective construct of internal order: better to be stressed than disorganized.

The constructs with the greatest intensity correspond to linked identity dimensions to performance ('tolerance to the uncertainty', '10 results', 'perfectionist'), which suggests that The system is primarily structured around self-efficacy and achievement. Emotional constructs They exhibit less intensity, which would support the hypothesis that discomfort operates as a consequence of the way in which these are preserved. nuclear dimensions.

	Intensity
No tolerancia - Tolerancia ince	0.376
Resultados norm - Resultados de	0.364
Chapuzas - Perfeccionistas	0.29
Control - No control	0.25
Desorganizada - Organizada	0.236
Vagas - Trabajadoras	0.218
Estresadas - Relajadas	0.183
Ineficiente - Eficiente	0.147

In the cluster analysis of the elements, the strong relationship between the 'ideal self' and 'Pastor', the latter being an important pillar to support and reaffirm the change since he represents that the change is not something abstract, it is plausible and embodied in someone close.



From the perspective of personal construct theory, this reduces the threat of change, because the new pole is not unknown.

A marked discrepancy between the current self and the ideal self, with, in addition, a

The distance of 3.33 was obtained from the element distance matrix. This distance is not evenly distributed among the constructs, but rather appears to be concentrated in specific dimensions related to performance and distress management. The greatest discrepancies are observed in the constructs 'stressed/relaxed' and 'normal results/10 results' (visual analysis of the Grid, first image), suggesting

that the gap between the two elements is mainly articulated around perceived self-efficacy and emotional regulation, rather than organizational aspects of the self."

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	(1) Yo - Actual	(2) Madre	(3) Hermana	(4) Pastor	(5) Yo - Ideal
(1) Yo - Actual	0.00	3.32	2.88	2.92	3.33
(2) Madre	3.32	0.00	1.97	3.11	3.24
(3) Hermana	2.88	1.97	0.00	2.63	3.05
(4) Pastor	2.92	3.11	2.63	0.00	1.62
(5) Yo - Ideal	3.33	3.24	3.05	1.62	0.00

A possible implicative dilemma arises, because while change mobilizes towards a
The 'non-controlling' aspect in daily life would imply being 'disorganized,' something undesirable because this is one of the congruent and identity-defining constructs of the self. This would be the kind of knot that perpetuates discomfort and hinders change.

Congruent	Discrepant	R
2. Organizada - Desorganizada	8. Control - No control	0.62



Being less controlling implies risking the construct of "organized," which is identity-based and consistent. And if "organized" is closely linked to "efficient" and "perfect results,"
So letting go of control isn't just about managing anxiety; it's about jeopardizing one's sense of competence. This leads us to hypothesize about whether stress is acting as a price to pay.
Paying to maintain identity consistency is costly, but it preserves the "organized/efficient/perfectionist" performance cluster. If control is reduced, the system risks sliding into "disorganized," which would be more threatening than stress itself. This makes malaise functional within the system and highly resistant to change.

In intervention, several possible strategies could be:

1. Address the implicative dilemma directly. For example, explore whether “less control” truly equates to “disorganization” in breaking the chain. This weakens rigid involvement.
2. Redefine success in terms of process and not just results.
3. Introduce a new node. If “organized” implies “control,” organization can be redefined by introducing, for example, the construct of flexibility. This allows the system to be expanded and enables changes without threat.
4. Introduce new dimensions that are not linked to performance (for example,

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perseverance, curiosity, learning).

5. Differentiate between “personal value” and “academic performance”.

The unease does not appear to be based solely on conflicting constructs such as
'stressed' or 'intolerance of uncertainty', but in the close link between these
and consistent core constructs such as 'organized' and 'efficient'. Stress could be functioning as a mechanism for maintaining identity coherence, in the
to what extent reducing control or organization would threaten core dimensions of the self.

Finally, the proposed development may be more extensive than usual; however, this extension has been deliberate, in order to actively work and become practically familiar with the different tools it offers.

Psychlab.

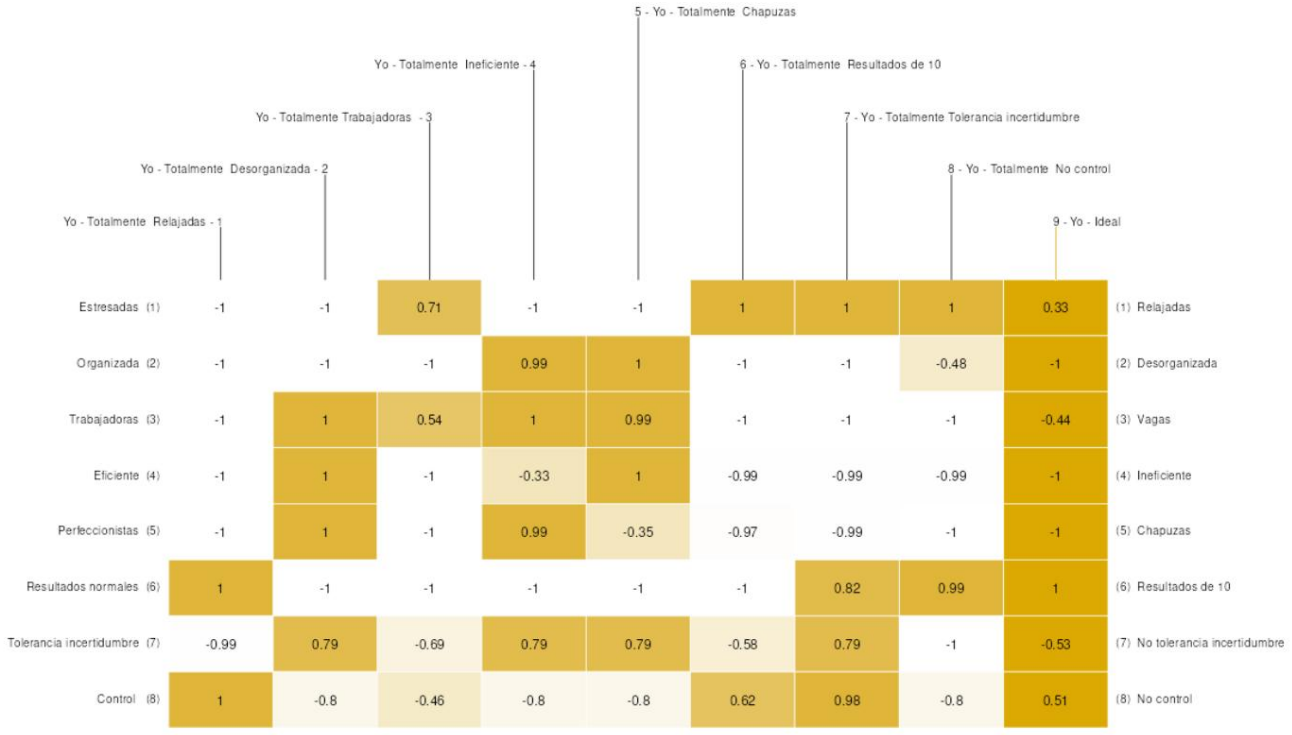
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III. Modeling the System of Personal Meanings

In this study, I will revisit the data collected in the previous Grid and use my personal case to explore the network of personal meanings and their relationships using the WimpGrid tool. Its main objective is to investigate how individuals anticipate consequences by generating changes in their self-perception. Therefore, the subject of this study is a 21-year-old university student who would like to change the stress she experiences in situations such as exams and those that make her doubt her own self-efficacy. After failing to manage this temporary stress, she notices that even after the stressful situation passes, it transforms into chronic anxiety that ends up dominating the range of coping strategies she might possess in situations where she previously acted effectively. To manage these situations, her coping mechanisms are usually based on worry, both conscious and unconscious, which creates limitations in her daily life. Despite trying different methods and achieving a greater sense of psychological relaxation, the presence of somatic symptoms persists across various situations.

Thus, the ultimate goal of the analysis will be to change the discomfort caused by stress, understanding that this means seeking to be more relaxed. Based on the previous Grid analysis, it can be considered that I have a self-reinforcing negative loop of control-organization-stress that currently helps me carry out my daily tasks, and therefore will be resistant to change despite generating this anxiety.

Below we can see the WimpGrid Data Table:



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The same modus operandi as in the Grid will be applied to the generation of the constructs, so it will be taken for granted.

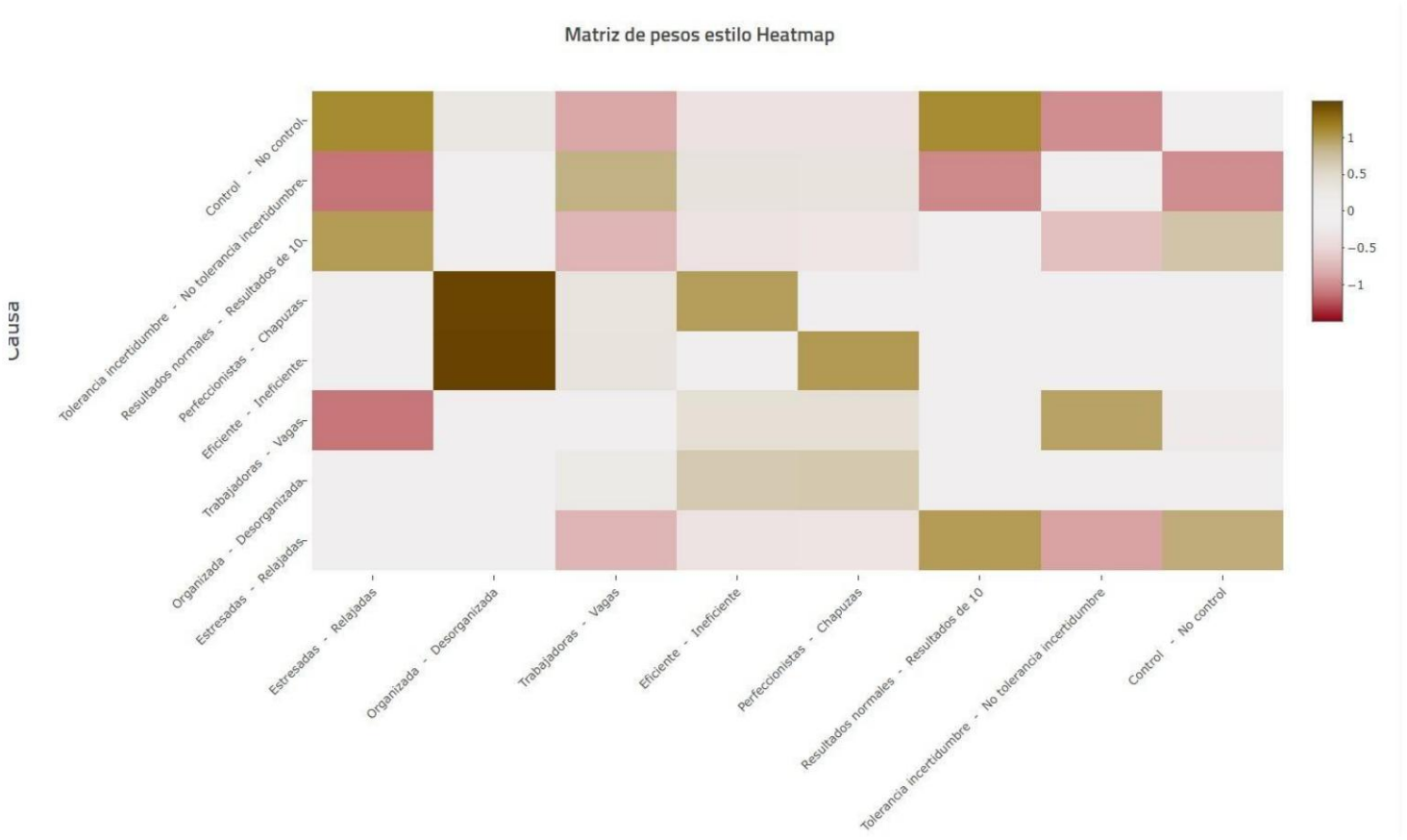
To conduct the interview, I followed the Psychlab guided form application protocol.
I chose this method because I already had a grid created and registered on the Psychlab platform, which facilitated the generation of the hypothetical self interviews, recording the scores in a direct and dynamic way. In this way, the hypothetical self-assessments will be automatically generated by the form according to the established criteria.

1. If the Actual Self is defined at one pole, your hypothetical self will be the most extreme score of the other pole.

2. If the Actual Self is not defined at either pole (intermediate score on the scale), but the Ideal Self is, we will use the most extreme score oriented in the same direction as the ideal.

3. If the Actual Self and the Ideal Self are not defined, the pole will be chosen arbitrarily.
construct right.

I would visualize the hypothetical situation before interacting with the slider in order to only move those sliders where I anticipated a change resulting from the situation. For example, when I imagined a situation where I was completely 'out of control' (not controlling, alert), some of the changes I anticipated were being more relaxed because I wouldn't be in alert mode all the time, and having a better tolerance for uncertainty because I wouldn't need to control or know everything that was happening, allowing me to flow more.

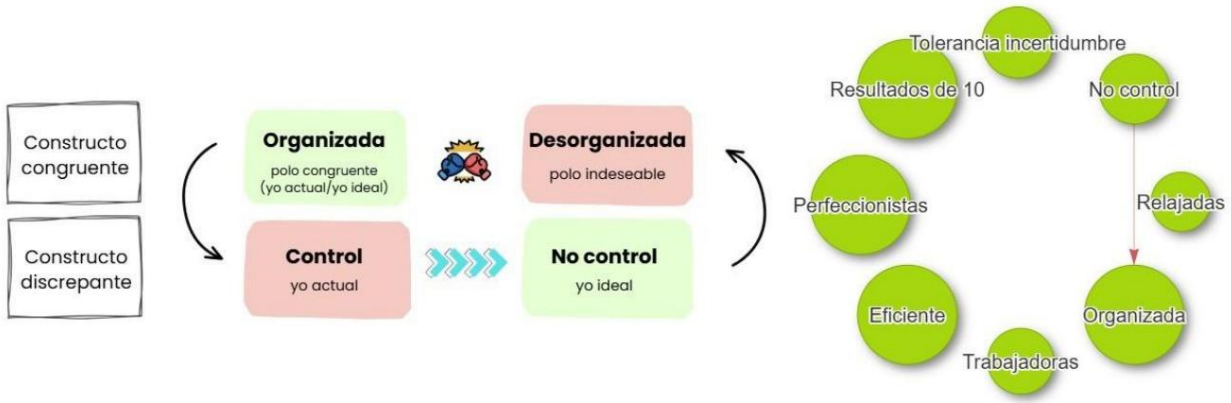


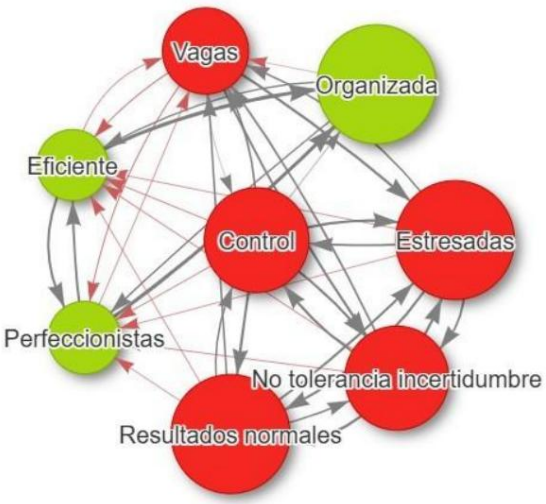
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Furthermore, this is an example that we can clearly see reflected in the Weight Matrix. We observe that changes towards being 'no control' have a strong implication both with 'relaxed', directly; and
with 'no tolerance for uncertainty', albeit indirectly, resulting in greater anticipation of the 'tolerance for uncertainty' pole.

By following these steps, I tried both to avoid generating indirect causal relationships, so as not to overestimate the effects later; and to avoid reversing the causality of the changes, that is, responding in such a way that the anticipated consequences preceded the causes (responding about what changes should have occurred to reach that hypothetical situation), since this situation would lead to inconsistent results.

Observing the Digraph of the Ideal, the Implicative Dilemma stands out again in the control and organization nodes mentioned in the Grid analysis.

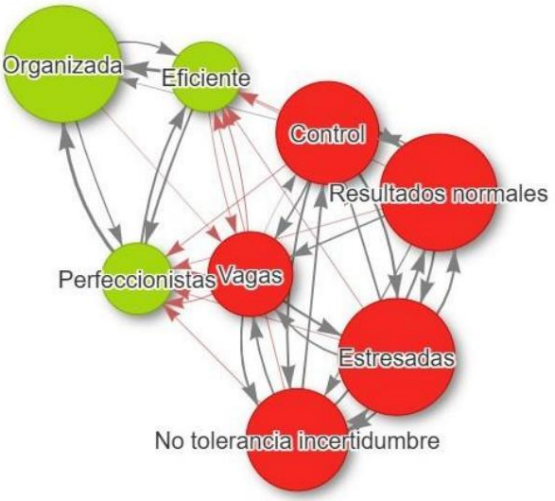




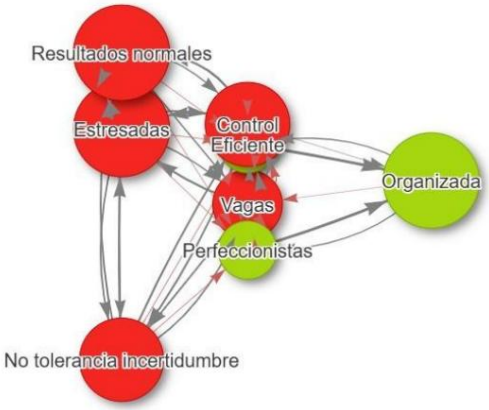
Regarding the analysis of the Autodigraph in the Reingold-Tilford algorithm (rtcircle *design)* based on the centrality and periphery of the system, we see how the control node takes an influential and “controlling” role in the network, being able to form a nuclear construct that is resistant to change due to the large amount of *outputs* it produces and the small amount of *inputs* it receives.

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On the other hand, evaluating the Graphopt algorithm, it is observe a positive feedback loop quite clear of the organized-perfectionist nodes-efficient and another feedback loop and remarkably rigid self-regulation of the poles negative.



On the other hand, multidimensional analysis *(MDS),* based on the similarity of the nodes, defined as the ability to cause the same effects in the system; allows us to visualize how being stressed and obtaining normal results can exert the same effects, influencing each other and the system to maintain the negative cycle that perpetuates discomfort.



The application of the WimpGrid interview and the subsequent analysis with Psychlab allowed me to understand how the Grid Technique translates subjective personal meanings into objective analyzable data. I learned to interpret graph patterns such as centrality, closeness, and feedback loops, This makes it easier to identify which dimensions are most central to a person's system of meanings. I also understood the importance of the proper generation of constructs and the elaboration of the situations of the Hypothetical Ego, since they directly influence the quality and depth of the analysis. This experience helped me connect theoretical concepts from personal construct theory with their practical application and its clinical interpretation.

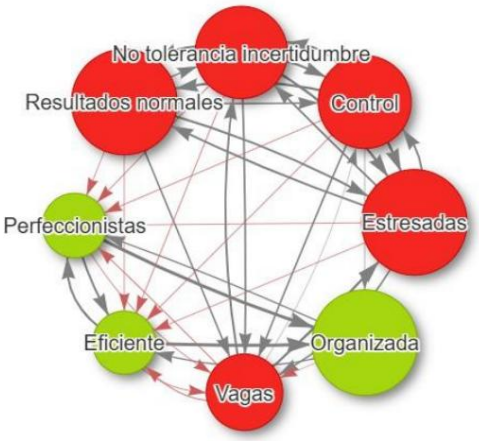
If I were to repeat the experience, I would pay more attention to the initial phase of obtaining constructs, ensuring that they are clear, truly causal, and relevant to the evaluation. Also I would dedicate more time to familiarizing myself with the different analysis options and indices that it offers. a program to make interpretations through new perspectives. Furthermore, it would complement the quantitative results with a deeper qualitative exploration, integrating personal meaning of the constructs to enrich the overall understanding of the cognitive system being evaluated.

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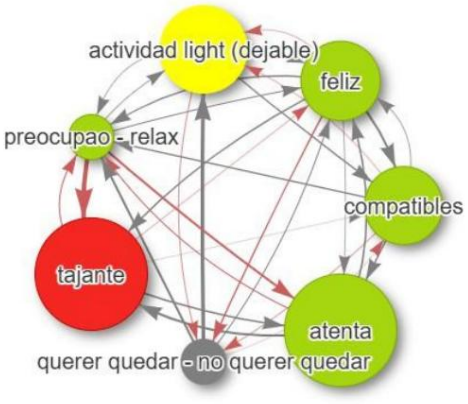
IV. Identification of psychological obstacles that hinder change

For the development of the different sections of this work, data from two different subjects will be used. This decision responds to the need to illustrate certain processes and characteristics that are more clearly manifested in one of the subjects and that
They could not be described with the same precision based on the other. Furthermore, the inclusion of both cases allows for a broader analytical perspective and provides a richer and more complete view of the phenomenon under study, incorporating nuanced differences that contribute to a better understanding. The subjects of the study will consist of myself and a person from my close circle, Elena. The selection of these two cases is based on the availability of the necessary information and the possibility of accessing their respective systems of personal meaning in detail to study the feasibility of change through Psychlab's graphical analysis.

The first subject, myself, is a university student of
A 21-year-old identifies her stress reaction to evaluative situations, such as exams, as problematic. When she fails to manage this stress in the moment, it doesn't disappear completely; instead, it persists.
evolves into a more lasting state of anxiety.



Her most frequent response to these situations is constant worry, both conscious and automatic. Although she has tried various strategies and has occasionally experienced some mental improvement, the physical manifestations of her distress continue to appear in different contexts.



The second case involves Elena, an 18-year-old in a relationship who says she'd like things to flow more naturally. She feels her partner is more relaxed and plans less.

so that the plans emerge as we go along.
This causes Elena, who identifies as a
A person who is 'decisive' rather than capable of 'flowing' may not feel entirely satisfied with the situation. It is worth noting that the elicited constructs will have their focus of convenience in this
in the context of a dating relationship, specifically showing its maximum usefulness.

To study the obstacles that may be hindering change, the first step will be to analyze
The Implicative Dilemma that arises from *my* RepGrid (from now on Lucia), since Elena does not manifest this event.

Thus, a possible implicative dilemma arises, since while change mobilizes towards

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A 'non-controlling' aspect in daily life would imply being 'disorganized,' something undesirable because this is one of the congruent and identity-defining constructs of the self. This would be the type of knot that perpetuates discomfort and hinders change, as we can see in the graph where each of its elements is described.

Congruent	Discrepant	R
2. Organizada - Desorganizada	8. Control - No control	0.62



Reducing 'control' involves jeopardizing the construct of 'organized,' which constitutes an important part of identity and with which one feels personal consistency. Since 'organized' is strongly linked to concepts like 'efficient' and 'achieving top results,' relinquishing control would not only affect anxiety regulation but also threaten self-perception.

as a competent person, in addition to compromising their actual performance. This suggests that stress
This could be functioning as a “price” to pay for maintaining identity coherence: although costly, it protects the set of characteristics associated with performance, efficiency, and perfection. If control is reduced, the system risks sliding into disorganization, which would be more threatening than stress itself. In this sense, discomfort is not simply a negative effect, but rather serves a function within the system, which explains its strong resistance to change.

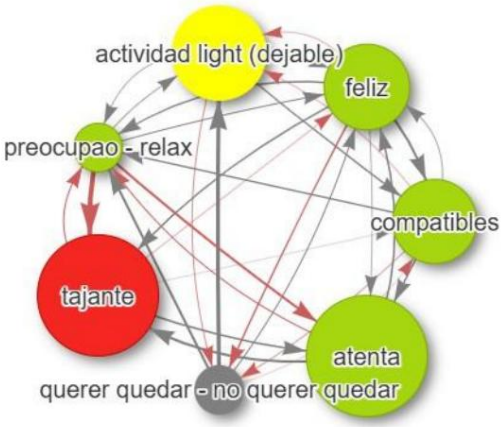
This Implicative Dilemma can be directly related to Inconsistencies in the Digraph of the Ideal, since by definition, an Implicative Dilemma is one in which, instead of achieving a congruent change of a construct (control) with the ideal (no control), another dimension (organized/ disorganized) that was already at the ideal pole is mobilized. In short, this situation gives



This gives rise to an indirect relationship between both constructs, producing the so-called Inconsistency of the Ideal, in which achieving one ideal mobilizes the opposite pole of another ideal that is also valued, as we observe in the Digraph of the Ideal.

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On the other hand, the exploratory graphic analysis of Elena's network of personal meanings shows us It allows you to perceive at a glance a series of inconsistencies, including:



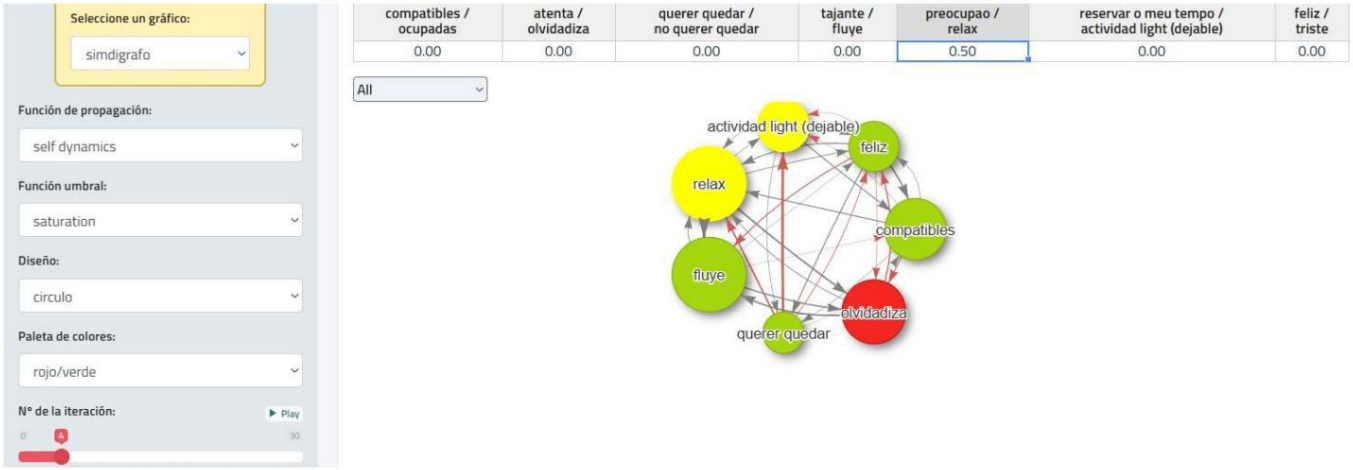
- The dilemma construct: 'activity 'light(lettable)/reserve your time', in which he places the ideal self at a score of 0, making it clear that he is not sure if the best One way to improve the situation is to do activities that are easy to put aside when new plans arise, or not to be influenced by your partner and reserve your time for your own interests. Primarily, it would be a lack of Motivational tension arises from uncertainty, although it's possible Elena is unable to choose the 'reserve her time' option for fear of negatively impacting the relationship, since she defines herself, both in her current self and her ideal, as 'compatible' (the opposite of 'busy'). This ambivalence may be interfering with the change, as it leads Elena to make initial attempts to prioritize her time, but these ultimately end up being set aside in favor of making plans together, which sends the wrong signal of availability to her partner.

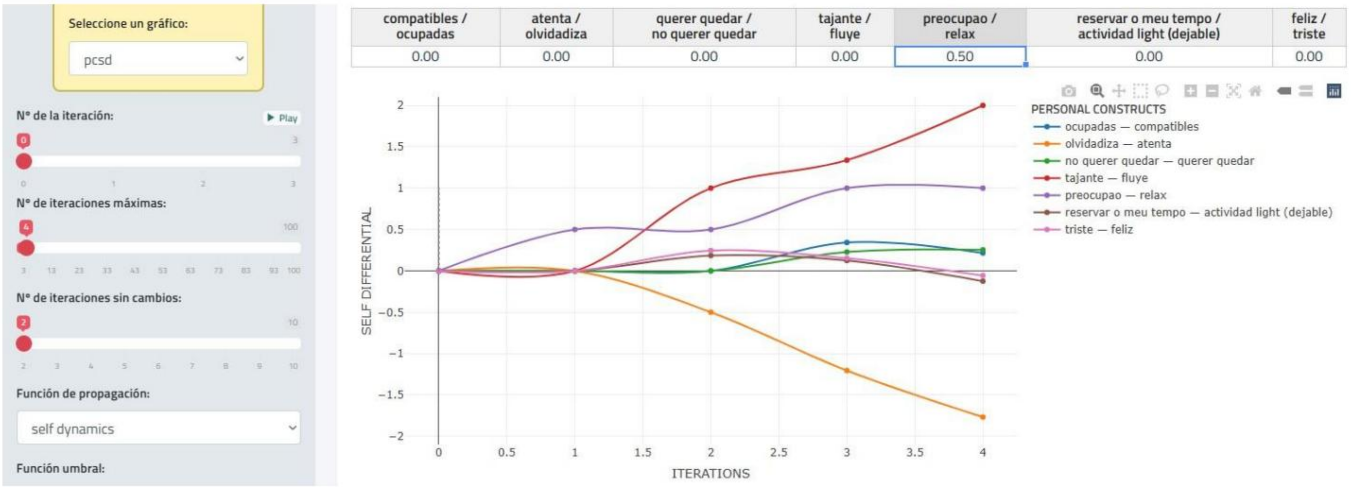
- Self-Disorientation: the construct 'wanting/not wanting to meet', which scores the current self as 0, while the ideal would be slightly negative (wanting to meet). As in the dilemma construct, it hinders the change process since it does not allow for charting a path towards improving the situation because there is no... a defined starting point. She perceives that previous action plans are not proving efficient, but the viability and performance of other new ones is still unknown Therefore, she doesn't dare to take a stand. These problematic constructs ('desirable activity' and 'wanting to meet') may be the manifestation of a turning point for Elena, generating a moment of profound change.

- On the other hand, the construct 'worried/relaxed' stands out, which, as our colleague Jose Manuel Roa Clemares commented in the forum for Topic 4: "Constructs dilemmas and constructs of self-disorientation." There are situations in which an intrinsically ambivalent construct emerges, in which a score of 0 both in the current self (Disorientation of the Self), as in the ideal (C. Dilemmatic). In these cases, the system, according to its definition, labels them as congruent. However, this does not mean they cease to be a relevant focus to explore in order to understand the network of personal meanings. As the tutorial team highlighted: *"Here, the interesting thing is to observe what effect this indefiniteness has on the rest of the system: whether it fulfills a stabilizing function or, on the contrary generates blockage and in which construct(s) ."* Following the proposed guidelines,

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I delved deeper into the situation using the lab tool that provides
Psychlab to assess what would happen if the position of the current self were changed in the construct.





• This first set of images represents a situation in which the construct is oriented towards the 'relax' pole (0,50). As we can see, in the fourth iteration the system plausibly redistributes itself and we can observe changes in the following constructs:

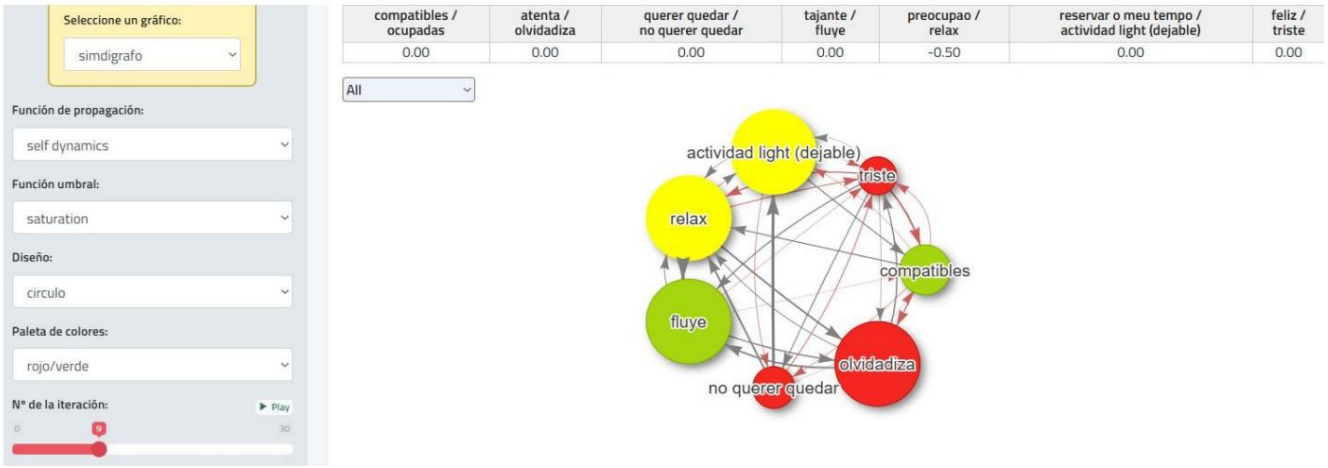
ÿ 'It flows': a notable positive progress is observed towards the pole of ideal.

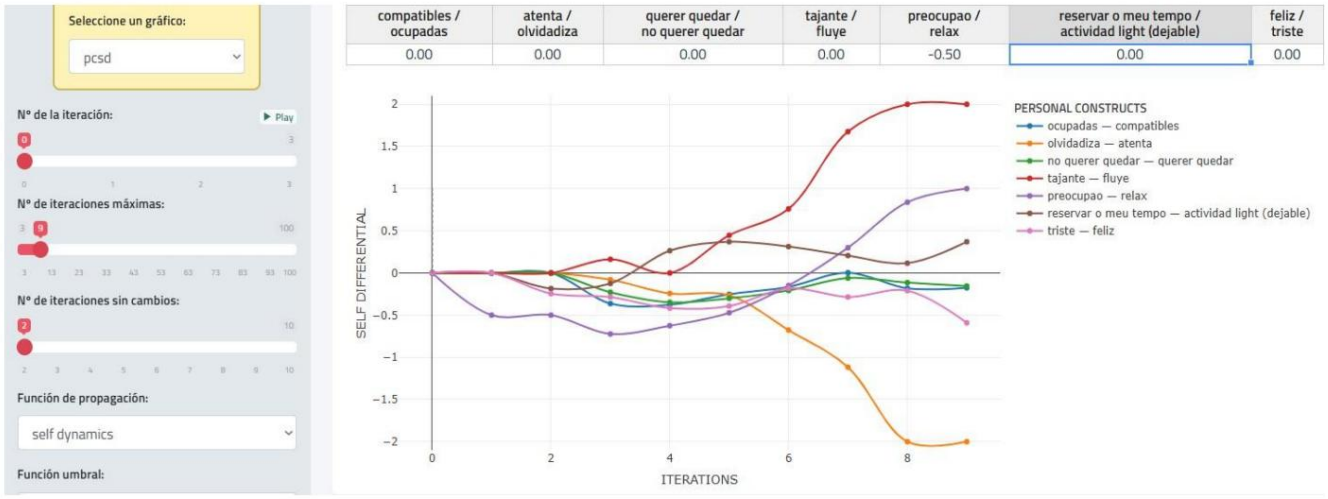
ÿ 'Wanting to stay': part of a starting position of Disorientation from the Self towards a mobilization to the congruent pole.

ÿ 'Forgetful': moves towards the negative pole. This situation It invites reflection on whether the effects of being less 'attentive' (despite being incongruous with Elena's identity) could constitute an area that, with an appropriate approach, offers opportunities for development and positive growth.

ÿ The rest of the constructs undergo superfluous variations.

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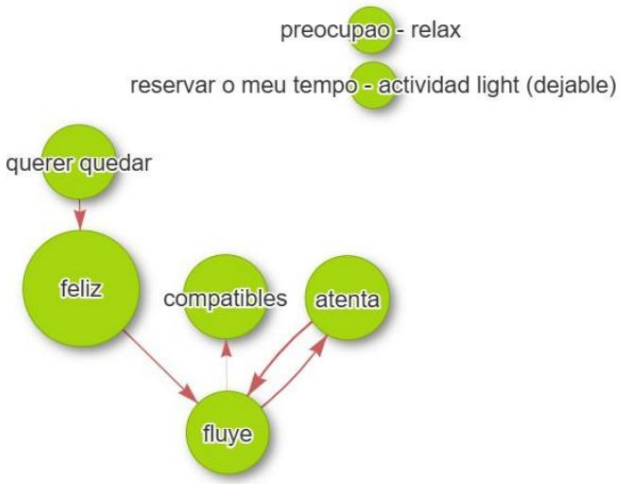


- The second set shows the construct at the 'concerned' pole (-0.50). If we compare the two images of *the symdigraph* we will see that, In essence, most changes are quite similar, with the exception of the constructs 'happy' and 'wanting to meet,' so a priori, this would appear to be the option with the highest cost for the subject. The situation is somewhat intangible, as it occurs in iteration 9; however, the following evolution of the construct network is observed:
  - ÿ 'Flow': again, it moves to the congruent pole.
  - ÿ 'Forgetful' and 'not wanting to stay': they move to the undesirable pole.
  - ÿ 'Sad': highlights that it positions itself in the undesirable pole, generating large costs for the system.

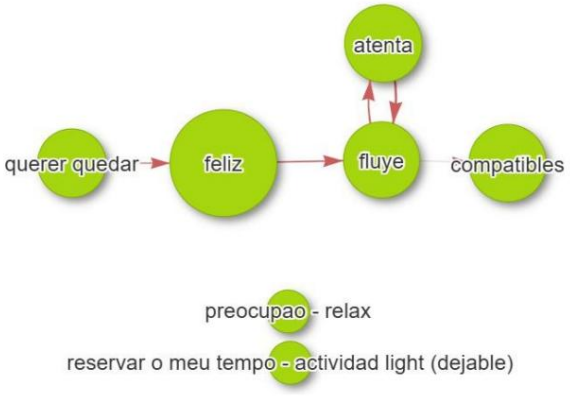
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Therefore, observing both interactions, it can be stated that it is not an isolated construct, but rather has the capacity to block and generate changes in the system; and specifically, in the aforementioned constructs. While it is true that the subject would not be clear about the anticipated consequences, since it is an ambivalent construct, we do see that by positioning themselves at the 'relax' pole, in addition to having rapid and plausible effects (iteration 4), they would also obtain a promising range of benefits, albeit at the cost of sacrificing 'attentiveness' from their identity profile.

To conclude the analysis of Elena's available phenomena, we address the inconsistencies in the Ideal Digraph. This graph allows us to identify potential incongruities between personal goals and ideals that may be causing blockages in the network. In this case, the network exhibits several inconsistencies. We could address the situation by intervening on the 'flow' construct, as it has the most inverse relationships, redefining it to be compatible with the other nodes.



On the other hand, the feedback loop that takes place in this digraph stands out, so it would be worthwhile to delve deeper into this situation to have a starting point in that the patient has more defined goals and greater internal coherence.



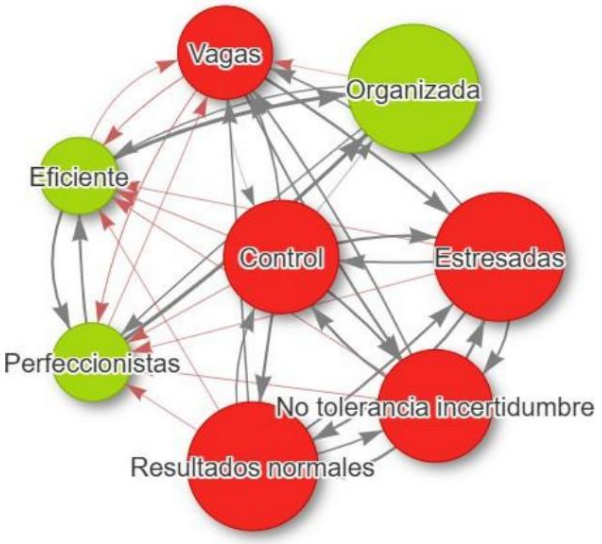
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All these ambivalences and inconsistencies within the system reinforce the hypothesis that we are at a turning point, a harbinger of profound change. While Elena doesn't feel entirely secure about new courses of action or changes to her identity, she is aware that her current situation is inconsistent and that this is generating costs rather than benefits. This may indicate that the system itself is "collapsing under its own weight," suggesting that internal tensions are inevitably leading it to readjust.

Systems of meaning seek coherence like planets seek a stable orbit; when they do not find it, they either change or break apart.

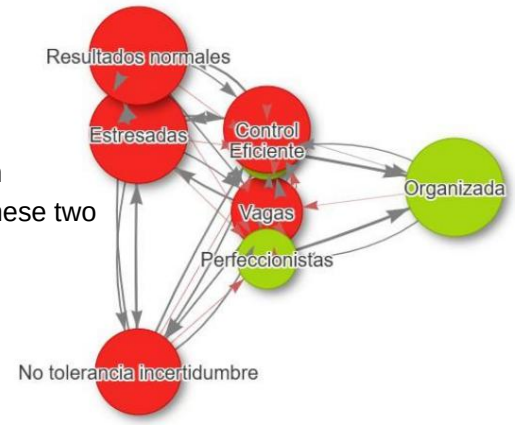
To further delve into the analysis of the system with the Fuzzy Cognitive Map and the study of the obstacles to change, we again take the information provided by the case of Lucia.

Regarding the analysis of the Autodigraph in the Reingold-Tilford algorithm (rtcircle *design)*
Based on the centrality and periphery of the system, we see how the control node takes on an influential and “controlling” role in the network, being able to form a core construct that is resistant to change due to the large amount of *outputs* it produces and the small amount of *inputs* it receives. It constitutes a construct that can be explored in greater depth to analyze in more detail its relationship with the system and its ability to unlock the network.



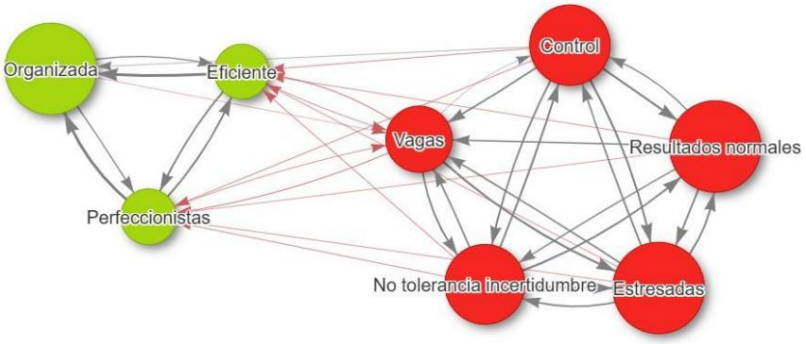
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On the other hand, multidimensional analysis *(MDS),* based on the similarity of nodes, defined as the capacity to cause the same effects in the system, allows us to visualize how being stressed and obtaining normal results can exert the same effects on the network, reinforcing each other and influencing the system to maintain the negative cycle that perpetuates the discomfort. These two These types of cycles, in turn, allow us to analyze three clear self-regulating loops:



Evaluating the *Graphopt* algorithm , two very marked types of feedback loops are observed:

- Positive feedback loop: direct relationship between its elements (gray edges).
- Negative feedback loop: inverse relationship between its elements (red edges).



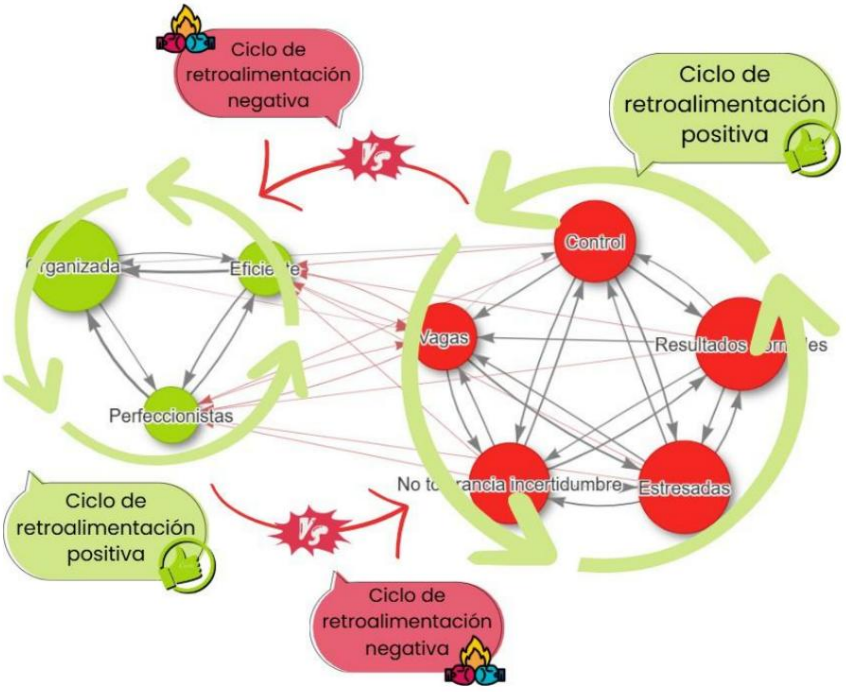
- Loop beneficial of Feedback (organized- positive perfectionist-efficient nodes): This direct nature of the relationship between its members provides a good support base for the system and preserves system coherence by representing “effective competition”. It also shows how the 'efficient' and 'perfectionist' nodes isolate and protect the core, 'organized' node from the influences of the conflicting loop.

- Conflicting positive feedback loop (vague nodes - do not tolerate) (uncertainty-stressed-normal results-control): a repetitive pattern that generates rigidity and circular self-justifications; however, it stands out because it functions as a protector against the threat to that “competence.” Despite this,
  It reinforces a maladaptive pattern, perpetuating discomfort and stress. It demonstrates that even patterns that generate discomfort can stabilize if they maintain internal consistency. It highlights that both loops, in addition to being connected by an outer circumference along their edges, are also intertwined.

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Yes, forming internal structures (triangle and pentagonal star), reflecting its deep feedback and rigidity.

- Superordinate negative feedback loop between the two loops of Opposite valence: it moderates an indirect relationship between the two subordinate loops. It brings dynamism to the system, encouraging it to relate as a whole rather than node by node; this reveals effects that would not otherwise be the case. observables by studying the different parts of the network separately. They function as regulatory forces that prevent one pole from completely dominating. In turn, this suggests a structural tension: the same system that sustains efficient identity contains the conditions that activate its fragility, generating a unique equilibrium.



From a clinical and global analytical perspective, growth would not imply eliminating one of the loops, but rather making the rigidity of their connections more flexible. When a system is sustained exclusively by two pillars (green loop and red loop), it loses adaptability, as it invests more energy in maintaining equilibrium and trying to prevent collapse than in adapting to face each situation.

Intervention strategies for the system could be:

1. Address the implicative dilemma directly. For example, explore whether “less control” truly equates to “disorganization” in breaking the chain. This weakens the rigid implication.
2. Redefine success in terms of process and not just results.
3. Introduce a new node. If “organized” implies “control,” organization can be redefined by introducing, for example, the construct of flexibility. This allows

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expand the system and enable changes without threat.

4. Introduce new dimensions that are not linked to performance (e.g., consistency, curiosity, learning).

5. Differentiate between “personal value” and “academic performance”.

Psychological health is not the absence of negative feedback loops, but rather the system's capacity to reconfigure itself without collapsing and to flexibly choose the most adaptive option for each moment. Ultimately, this diagram doesn't speak of nodes, but of dynamics. It doesn't describe what Lucía is like, but how she maintains that identity. And every system that...
It is maintained but can also be transformed, because psychological stability is not a fixed property, but an active process of self-regulation.

V. Case formulation and intervention strategies

Let's briefly recap the case of Lucía: a 21-year-old university student who identifies her stress reaction to evaluative situations, such as exams, as problematic.
When she fails to manage this stress in the moment, it doesn't disappear completely, but rather evolves into a more persistent state of anxiety. Her most frequent response to these situations is constant worry, both consciously and automatically. Although she has tried various strategies and has occasionally experienced some mental improvement, the physical manifestations of
Discomfort continues to appear in different contexts.

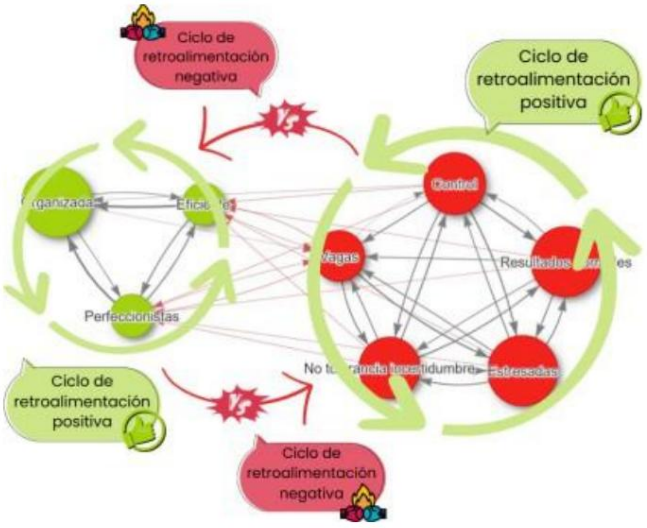
- The RepGrid is particularly useful for studying the subject's SSP and its constructs. The Implicative Dilemma stands out here: reducing 'control' implies
  This puts into play the construct of 'organized,' which constitutes an important part of identity and with which there is personal consistency. This suggests that stress might be functioning as a "price" to pay to maintain identity consistency.



- With the WimpGrid we analyze how the system behaves as a whole.
  This allows us to specifically visualize the dynamic relationships between the different nodes and sets of nodes. In this network we see how the nodes of

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Each valency reinforces the other, while the loops interact.
among themselves to try to give coherence to the identity and adapt to the context.



Narrative formulation of the case

a. History of the problem

i. Somatic symptoms: physical discomfort associated with times of stress
that affect their daily functioning.

ii. Uncontrollable appearance of discomfort: despite trying various strategies,
none seem to solve the problem.

iii. Reduced performance: being nervous prevents her from giving her best.
100%.

iv. Feeling of inadequacy: she feels insecure about her abilities, which affects her
emotional well-being.

b. Predisposing factors

i. Family expectations: their parents have always emphasized the importance
of good performance and good results.

ii. Internal model: she sees herself as a person who is disturbed by
ease in different situations.

iii. Own expectations: expects to be able to meet the expectations of teachers
and family members place their trust in it, even if it seems excessive.

iv. Personal vulnerability: success in terms of results and personal value equal to
academic value, limiting access to dimensions that could favor self-concept.

v. Lack of contact with students in equivalent conditions: this lack of contact with
peers in the same situation limits access to

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interpersonal validation processes and experiential contrast, potentially relevant for the reaffirmation of a favorable self-assessment.

c. Triggering factors

i. Rigidity and paternal perfectionism: his academic trajectory was Marked by the constant pressure and expectations of her parents, she developed a self-perception based on results. This also led her to prioritize planning and control from a young age in order to achieve outcomes, resulting in a rigid, controlling approach.

ii. Emphasis on the role of exams: from childhood exams have assumed critical all-or-nothing situations.

d. Maintenance factors

i. Ego-dystonic problem: although stress generates discomfort and rejection, She also believes that to some extent it is necessary to stay "vigilant".

ii. Somatic symptoms: can trigger a cycle of worry.

e. Reconstruction process

i. Therapeutic objectives

ii. Reduce the gap between your current self and your ideal self to improve self-concept.

iii. Introduce new dimensions that are not linked to performance (results).

iv. Redefine success in terms of process and not just results.

v. Differentiate between “personal value” and “academic performance”.

vi. Redefine the internal (nervous) model.

vii.Actual fit between expectations and the ability to meet them.

f. Strategies

i. Conversations with peers in the same situation. (2,3,4)

ii. Deactivation of alert state: let the concerns. (1,2,4,6)

iii. Highlight progress to validate the change process. (1)

iv. Emphasis on their own well-being (6)

v. Turn insecurity into effort (1,5)

g. Techniques and tools

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i. Emotional psychoeducation.

ii. Capacity analysis and setting limits.

iii. Psychlab.

iv. Mindfulness.

v. Sport.

h. Resources

i. Relationship with Pastor: social support from a close student friend, Lucía feels understood and sufficient.

ii. Self-care: maintain a healthy lifestyle, with good Nutrition and exercise. She highly values living in harmony and cultivating the body, mind, and spirit.

iii. Self-efficacy: considers that their insecurity is not based on real foundations but that it could be a result of their constantly changing expectations increase.

iv. External assessment: Lucía is highly valued by her environment

i. Motivation

i. Work capacity and perseverance: trusts in their abilities to to achieve the desired therapeutic change.

ii. Self-care: actively seeks personal well-being and strengthens self-esteem.

iii. Motivation for change: stress is acting as a burden that wants to leave it behind.

j. Difficulties

i. Functional belief about stress: believes that she needs to feel stressed in order to feel involved.

ii. Somatic symptoms: they remind her that the stress is still there even though she feels better.

iii. Academic results: she does not feel confident about whether her performance in the grade is sufficient.

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Therapeutic Feedback Report: Narrative Feedback



- **Identity**

We see that it's important for you to feel at peace with yourself.
You also value high performance through good
organization.

- **Problem**

You want to stop feeling so *anxious* and be able to cope with stressful situations more confidently.
(relaxed).

However, you mention that being *organized* and in *control* is important to you, which sometimes leads to an unwanted level of *stress* . On the other hand, you describe yourself as someone who highly values *efficiency,* organization , and achieving *good results.* This, however, sometimes leads you to be overly demanding *(a perfectionist)* and to resort to tools *(control)* that, in the long run, are not sustainable. The *dilemma* that arises is that achieving your desired changes would distance you from your sense of organization.



- **Redefinition**

What do you think about trying a different method to evaluate your performance?

You might realize that there are many other incredibly valuable factors that you're not currently considering. Following your desire to reduce *stress,* and respecting your *organization,* I invite you to broaden the scale you use to measure your personal worth, enriching it with many other dimensions that will lead you to a new level of personal well-being.

- **Work agreement**

You consider yourself a health-conscious person and want to feel good both physically and mentally. Therefore, I encourage you to draw upon your work ethic *(you're a hard worker)* to wage an internal battle, not because of other people's expectations, but by yourself, against that which is limiting your *results (stress, intolerance of uncertainty, control,* laziness).

Control should be seen as a tool (attention and knowledge), not the end in itself, with which to "cross the murky waters" of the student stage and achieve a life of well-being .

Did we reach that working agreement?

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“Be Like Water” Action Plan for the patient

Over the next few weeks, you'll explore new dimensions of performance and act as someone with complete confidence in your abilities, someone who doesn't need to meet everyone else's expectations to be successful. In your daily life, you'll try to feel more relaxed and let go of worries that you know don't serve you, because you know what you're capable of.

Be aware of when your "justifications" for not completing the proposed tasks are activated. Nothing is softer or more flexible than water, but nothing can withstand it.

Be like water.

Scales for monitoring the process

As a therapist-patient in this endeavor, I want to emphasize that the in-depth study and analysis of the network of constructs, as well as the introspective work, have proven to be an intervention strategy in themselves. In the days leading up to this follow-up, I have noticed the following progress, which could be considered as monitoring scales of the progress:

- Colleagues are perceived as a threat to personal worth ÿ They are perceived as figures with much more experience and practice from whom to learn.
- Demotivating insecurity ÿ Use that insecurity as work motivation.
- Grade points as the sole measure of competence ÿ Other measures such as actual performance (e.g., these tasks).
- Success in terms of results ÿ Perception of effort and intention.
- Errors as identity rather than behavior ÿ Errors are learning opportunities.
- Exclusive dependence on the expectations of others ÿ Self-critical judgment of performance.
- Constant concern about being enough ÿ Perception of secure self-efficacy that is not permeable to passing events.
- Unrealistic expectations ÿ I adopt an attitude oriented towards self-improvement I continue to reinvent myself by redefining my limits and exploring new development opportunities.

With a patient approach, I feel I'm improving in various aspects of my personal and academic life. I recognize that feeling a little stressed from time to time isn't bad; what's bad is the incorrect and excessive management of that stress.

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Final reflection

From a clinical perspective, the case formulation process has allowed me to achieve a deeper and more structured understanding of the subject's personal meaning system by complementing the analysis of the RepGrid's construct content with an exploration of its organization and relational dynamics in the WimpGrid. In particular, the RepGrid facilitated the identification of core constructs involved in the problem, as well as the central implicative dilemma, revealing that reduced control could be experienced as a threat to the construct of "organized," which is closely linked to her identity.

This finding proved particularly relevant, as it allowed for an understanding of stress not merely as a dysfunctional symptom, but as a coherent element within the system, serving to maintain identity consistency. For its part, the WimpGrid provided a systemic perspective that allowed observation of how these constructs are organized into networks of mutual influence, reinforcing the contingency between academic performance and self-worth, and thus contributing to the maintenance of distress.

Among the most useful elements for the formulation was the possibility of identifying particularly relevant dimensions through the analysis of factors (predisposing, triggering, and maintaining), such as the centrality of control, the equivalence between personal and academic value, and its contingency with personal validation. Likewise, initially less explicit constructs emerged, such as the importance of self-care to Lucía and her expectations of others, which provide new avenues for understanding the problem and guiding the intervention. In this sense, the case formulation proved especially valuable by providing an integrative framework that allowed this information to be organized in a coherent and clinically meaningful way, facilitating both the understanding of the problem and its feedback to the individual.

One of the main challenges was translating this inner complexity into an intervention proposal that respected the individual's identity, avoiding changes that could directly threaten core identity constructs. This meant guiding the process toward making the system of constructs more flexible and expansive, rather than eliminating them, fostering a distinction between personal worth and academic performance, and promoting the incorporation of new dimensions of self-esteem.

In general terms, the subject's system of personal meanings exhibits high coherence, although there is also some rigidity regarding performance as the primary criterion for self-evaluation. Nevertheless, a significant potential for change is observed, reflected in their capacity for introspection, their motivation toward well-being, and the presence of relevant personal resources, such as self-care, perseverance, and social support. These strengths, along with the establishment of a therapeutic alliance based on a collaborative model, are key elements for facilitating the reconstruction process.

I find it especially important to highlight the value of the collaborative model of constructivist epistemology in relation to the therapeutic alliance and the dropout rate.
Recent meta-analytic studies have found a consistent moderate correlation.

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There is an inverse relationship between the quality of the therapeutic alliance and the dropout rate in psychotherapy: a weak therapeutic alliance is consistently associated with a higher likelihood of premature treatment discontinuation. Meta-analytic evidence shows that patients with a weaker therapeutic alliance have a greater tendency to drop out of therapy, with a moderate effect size (Sharf et al., 2010). Factors such as the patient's initial motivation and the therapist's ability to establish and maintain a collaborative relationship also influence retention, linking alliance strength to treatment success. This means that therapeutic alliance explains a significant portion of the variability in therapeutic outcomes (Roos & Werbart, 2013; Jankowsky et al., 2024).

Taken together, this experience highlights the value of clinical formulation as a tool not only for understanding the problem, but also for identifying relevant constructs and precisely guiding the intervention, further encouraging the subject to become aware of the meanings involved in their distress and to actively participate in their change process.

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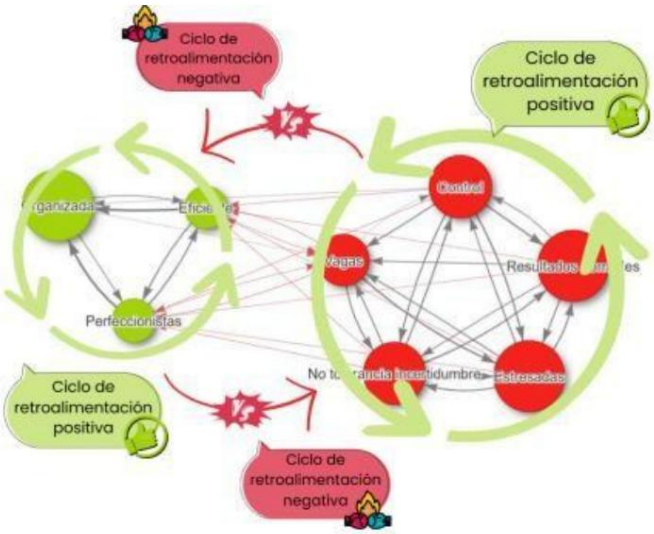
VI. Monitoring the change process

Let's briefly revisit the case of Lucía: a 21-year-old university student who identifies her stress reaction to evaluative situations, such as exams, as problematic. When she fails to manage this stress in the moment, it doesn't disappear completely, but rather evolves into a more persistent state of anxiety.
Their most frequent response to these situations is constant worry, both consciously and unconsciously.
as if automatically. Although she has tried to apply different strategies and has sometimes perceived some improvement at a mental level, the physical manifestations of the discomfort continue to appear in different contexts.

- The RepGrid is particularly useful for studying the subject's SSP and its constructs. The Implicative Dilemma stands out here: reducing 'control' involves putting the construct of 'organized' into play, which
  It constitutes an important part of one's identity and with which there is personal consistency. This suggests that Stress could be functioning as a "price" to pay to maintain identity coherence.



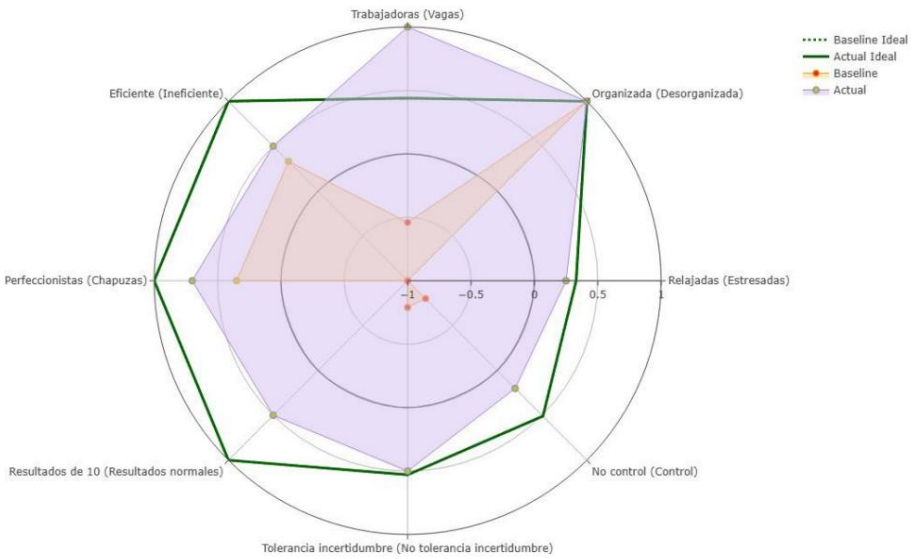
- With the WimpGrid, we analyze how the system behaves as a whole. This allows us to specifically visualize the dynamic relationships between the different nodes and sets of nodes.
  In this network we see how the nodes of each valency reinforce each other, as do the loops
  They interact with each other to try to give coherence to the identity and adapt to the context.



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In the therapeutic process, it is understandable that the patient wishes to deepen their understanding of what is happening to them. However, this inclination can pose certain limitations within the framework of categorical diagnosis, insofar as the adoption of a label can encourage forms of self-identification.
rigid and potentially interfere with the change process. In contrast, within monitoring through
In PsychLab, this same arrangement takes on particular importance, as it allows for the channeling of the
a natural tendency towards self-knowledge through the systematic observation of one's own evolution, favoring a
active and bilateral participation in the therapeutic process.

The Post assessment presents the changes produced after approximately two weeks of interval, showing a notable improvement in most areas.



As we see in the Ideal Adjustment graph, the constructs have shifted towards the Ideal Self pole, some
even reaching or exceeding it, as we see in 'hardworking', 'organized', and 'tolerance for uncertainty', and to a lesser extent,
'relaxed'. Others, such as 'perfectionist' or 'efficient', already showed a high value initially, because
They do show progress, but within a certain stability. Note that some constructs such as
'Perfectionists' or '10-out-of-10 results' show extreme values of the Ideal value, so it would be worthwhile to reassess whether these goals are based on realistic expectations. The remaining constructs have not lagged behind, but rather show
a notable advance with respect to the baseline (LB), for example, constructs that started from the negative pole
('results of 10') or that were very close to the value 0 ('no control') would increase to levels
close to 0.5.

Regarding the Adjustment analysis using the SSI index, the subject initially showed an index of 0.38 (in
ideal adjustment situation) which in the Post evaluation progresses until it reaches the value 1.

- The first graph (pre-measurement) indicates that there would be a similarity in terms of the subject's analysis of

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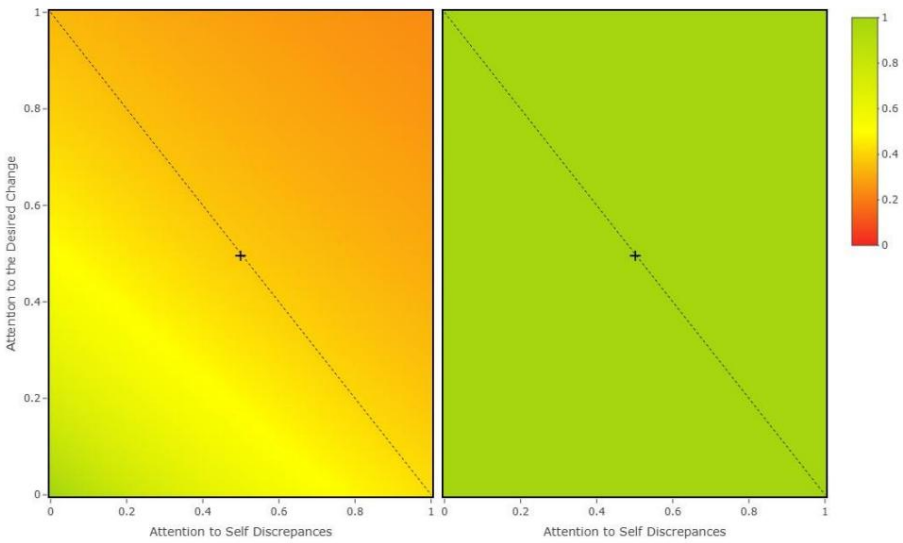
Discrepancies exist within their sense of self and their goals for change, showing only a slight tendency on the Beta axis to have a more negative impact than the Alpha axis. The low SSI level could indicate an initial phase of reorganization within the system. It could also represent a low degree of self-perceived alignment with the Ideal, or a high level of self-demand represented by goals with a relatively high aspirational status.

- We see in the second graph (post-measurement) that the system now shows a high degree of alignment with the Ideal. Such a high SSI may also represent a lower level of internal conflict, greater cohesion, and a high level of emotional stability across the subject's various evaluation possibilities.

As a point of reflection, in my opinion, this type of analysis could also be considered as a possible general measure of the subject's optimism/pessimism. The greater the valuation of the desire for change (Beta)

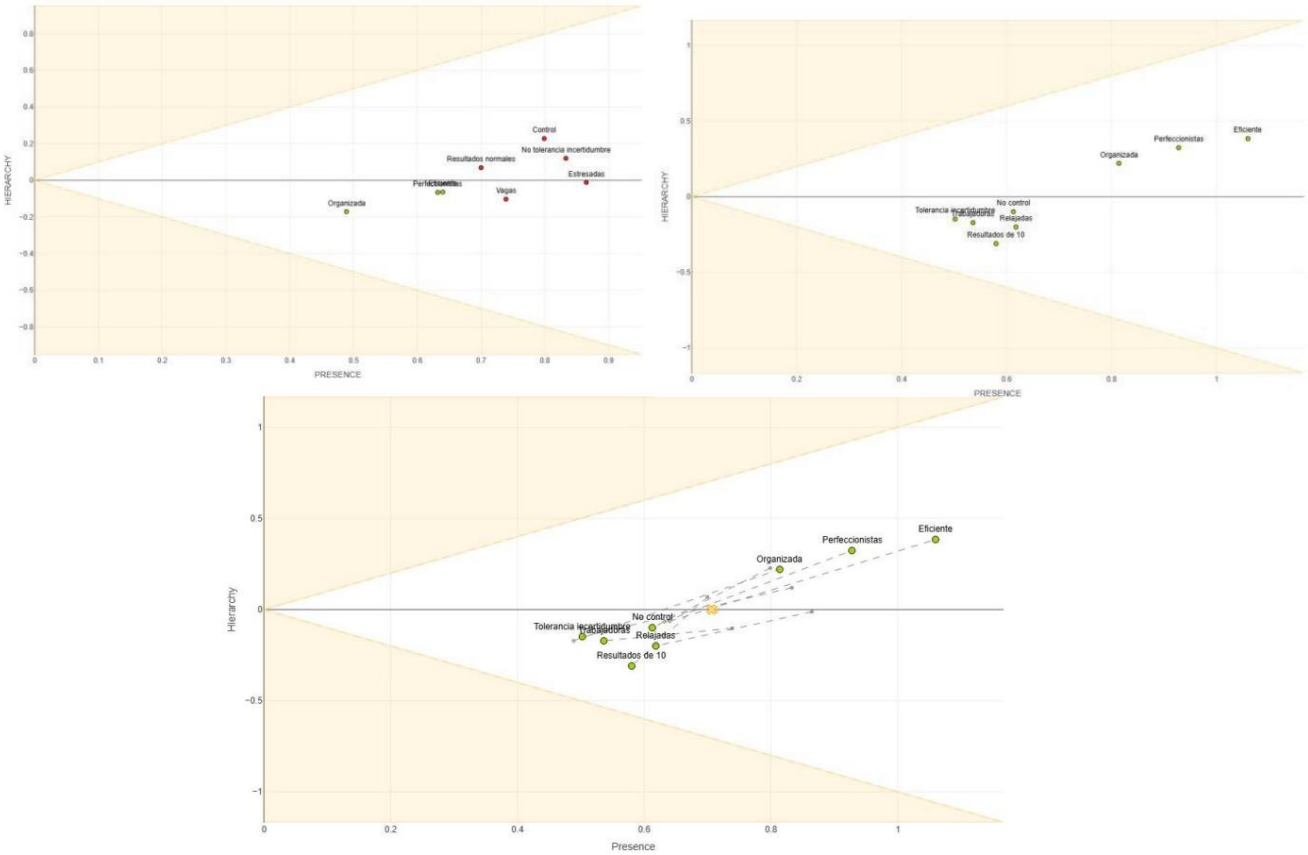
This would be consistent with a higher optimism indicator, focusing more on similar aspects than

This relates to the Ideal *(Ideal-centered style)* and to action towards change. On the other hand, a high valuation of discrepancies in the Self would be a greater indicator of pessimism; this manifests a tendency to focus more on what is not yet possessed and on the journey remaining to the Ideal, running the risk of falling into a negative loop of emotional burden from "what I don't have" *(current-centered style).*



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If we perform a centrality analysis based on the PB index, we see:



- In the first graph (Pre), considering the orthogonal dimensions of Presence and Balance of Involvement, we see how the construct 'stressed' was highly prominent (high presence), as were most of the negative constructs (control, lazy, etc.), while the positive ones were the least relevant in this respect. Regarding Balance, once again, the negative constructs stand out due to their
  great influence (positive balance) in the system and the positives as subordinate to these negatives (Negative balance).
- In the second graph (Post) we see how when the constructs 'efficient', 'perfectionist' and 'organized' are postulated as the main drivers of the system (both in Presence and Balance)
  (positive), this reorganizes in such a way that the network remains in the Ideal.
- In the third and final graph (Pre-Post monitoring), we observe the evolution of the system and how the negative constructs that were structurally very relevant at the beginning become secondary (although their polarity changes), and vice versa. The 'organized' construct stands out, going from being highly influenced (Negative Balance) and with the lowest Presence, to being one of those that now organize the system (mentioned in the previous graph).

Monitoring the PB Index allows for the identification of changes in both the presence and the Balance of Involvement of the constructs, demonstrating that the therapeutic process is not solely based on
the isolated change in the valency of each construct separately. On the contrary, it reveals its

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systemic and dynamic character, in which the constructs function as an organized whole,
influencing each other and reorganizing themselves according to their interrelationships. This is why in the
The Post graph does not show the constructs in the same position as the Pre graph, but with positive valence; instead, they also
need to modify their relationship with the network of personal meanings, generating changes
with respect to its initial position.

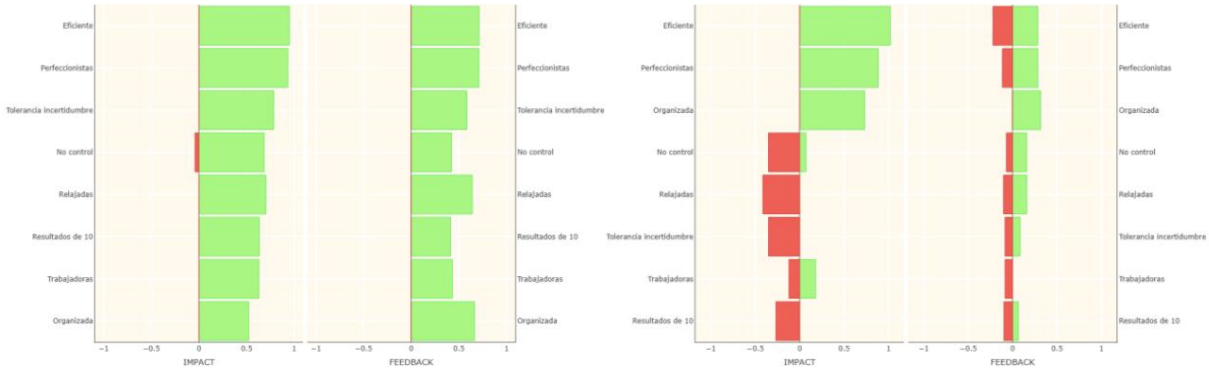
The Change Implications analysis allows us to evaluate the assessments of the different constructs
that the system generates as a result of the subject's anticipations.

- In the Pre (left) section, we interestingly see how the system places all constructs in the green quadrant (Positive Impact and Feedback), predicting beneficial changes that will bring us closer to defining the Ideal. The more extreme constructs, 'perfectionist' and 'efficient,' stand out as the main agents of change. This supports the prominent position these two constructs held in the PB-Post graph.

- In the Post (right), the first thing that stands out is the green quadrant, where 'organized', 'perfectionist', and 'efficient' are concentrated. These constructs combine high positive impact and high positive feedback, making them true driving forces of the system. In terms
  Clinically, this indicates that moving towards these poles not only brings us directly closer to the ideal, but also
  It facilitates a coherent global reorganization. Especially Organized, which presents the most feedback, suggests a construct with strong structuring capacity. In contrast, 'hardworking'
  It appears to have a slightly positive impact, but negative feedback. This implies that, although moving in this direction brings us closer to the ideal in direct terms, it generates tensions or imbalances in the rest of the
  system. The upper left quadrant, where 'relaxed' and 'no control' are located, shows constructs
  with negative impact, but positive feedback. This means that moving towards them would move us away from the ideal.
  explicit, but at the same time it would facilitate some systemic flexibility. This is a clinically enriching finding, as it suggests that dimensions initially incongruent with the ideal could play a role in facilitating global change. Finally, the lower left quadrant,
  Where 'results of 10' appears, it reflects the most unfavorable position: negative impact and feedback
  Negative. This indicates that changes towards this construct would have negative consequences for the system's adaptive reorganization. Its proximity to 'uncertainty tolerance', with values
  close to zero also suggests an ambiguous or unstable position within the global structure.

Regarding the local analysis of the constructs, we see at a more detailed level than in the Pre analysis (left),
In general, it is again seen that positive changes are anticipated both in terms of synchronization with the Ideal Self and with
The system feedback. On the other hand, regarding the Post graph (right) we again see relatively low data
powerful impact factors, as well as slight variations in the level of feedback anticipated by the network. These
These analyses could be interesting at a clinical level to decide which constructs to address first according to their level of Impact
and Feedback in the SSP, with those in which changes are predicted being the most relevant.
beneficial and also supported by a solid value in the indices.

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Monitoring the change shows signs of a movement towards a more integrated and aligned state
With the Ideal, this is not so much due to a direct increase in control-related behaviors, but rather to a reorganization of the meaning attributed to the constructs involved. In this sense, it is especially relevant that the decreased need for control and the greater tolerance for uncertainty have not resulted in a
loss of organization, but rather, on the contrary, have facilitated the experience of feeling more organized a less rigid and demanding form. This shift suggests that the construct of 'organized' is beginning to detach itself from the requirement of strict control, allowing for a more flexible form of organization and
consistent with the Ideal.

Overall, the system appears to have gained in coherence, insofar as some constructs that
They previously functioned in a more rigid or strained way, but now they begin to reorganize themselves more harmonious. This change does not imply merely an isolated modification, but a transformation in the relationship between the constructs, favoring a more integrated configuration. However, this reorganization
It also opens up the possibility of new structural contradictions. For example, constructs' outcomes
The terms '10' or 'hardworking' could remain potentially strained dimensions if they continue to be associated with self-demand or sustained effort as the main path to personal validation, without being integrated
fully in this new, more flexible balance.

Among the most significant changes, therefore, is the redefinition of the control-organization axis, which
It shifts from being based on rigidity to relying on greater confidence in one's own ability to function. Likewise, the strengthening of constructs such as 'organized' or 'efficient', accompanied
A less centralized control points to a hierarchical reorganization of the system more aligned with the Ideal.

Based on these results, the consolidation of an identity can be proposed as a future working hypothesis.
organized and not dependent on strict control, as well as exploring the role of the construct 'perfect 10 results' and its possible link to self-demand. Similarly, it might be relevant to continue
deepening the integration of 'tolerance for uncertainty' as a resource compatible with effective functioning, thus promoting a more flexible, coherent and sustainable system of meanings.

The observed process points to a modification in the way the system is structured, favoring a more balanced operation that is less dependent on previous restrictive dynamics. Thus, the

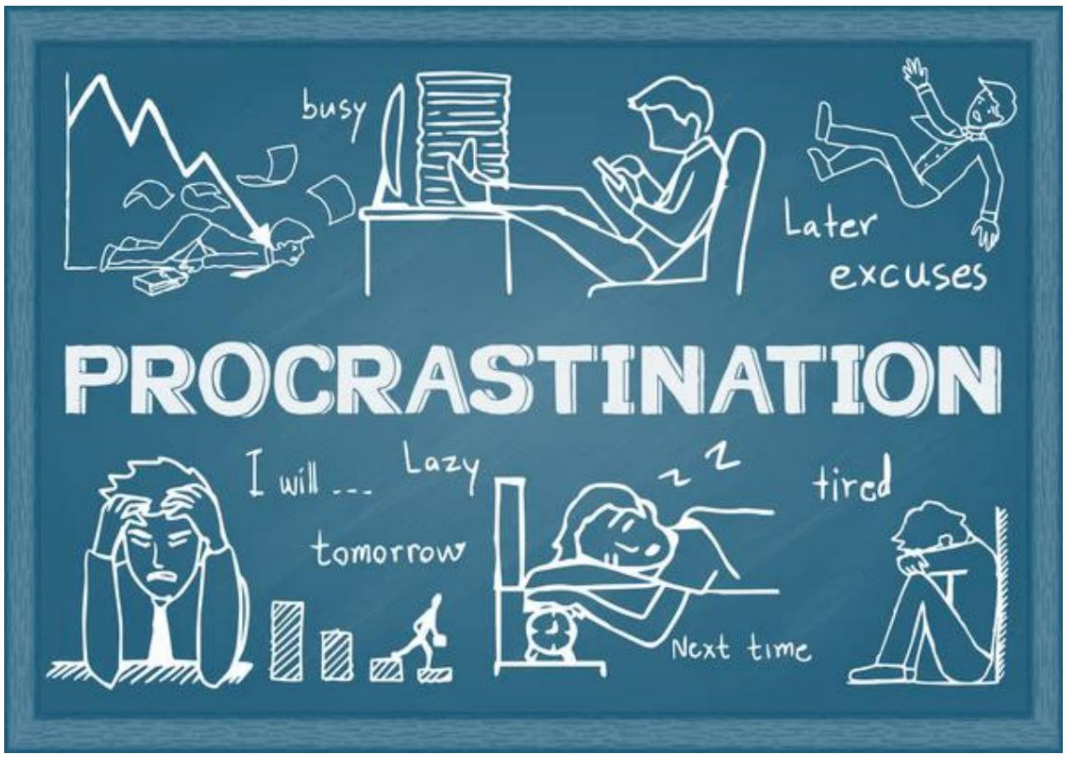
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Change is not only reflected in the valence of the constructs, but also in the relationship they maintain with each other and in the place they occupy within the global organization.

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APPENDIX III: Research Project on Procrastination, Personality and the System of Personal Constructs

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Research design on procrastination,

Personality and the System of Constructs

Personal

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Influence of temperament and identity on modifying procrastination
technology

Summary

Research on technological procrastination has tended to focus on the
self-regulation, often overlooking the individual's idiosyncratic structure of meanings.
This research project would propose to evaluate the impact of a digital hygiene intervention
integrating Cloninger's nomothetic approach to personality traits and the perspective
idiographic of Personal Construct Theory. For the method, a quasi-
Longitudinal experimental study with pre- and post-intervention measures. Participants would complete
Initially, the TCI-R-67 was used to establish stable personality profiles, and they would draft a
Self-characterization, whose constructs would be evaluated using the WimpGrid interview to map
their Personal Meaning System. They would then undergo a behavioral intervention.
of 14 days of digital hygiene. Finally, their cognitive structure would be reassessed using a post-test
with the WimpGrid. The expected results would indicate that the intervention would produce an improvement
behavioral changes in procrastination markers. At a structural level, a change in behavior would be expected in procrastination markers.
topological restructuring in the mathematical indices produced by the PsychLab software.
The findings would demonstrate that personalizing psychological interventions would be effective
key to modifying digital habits deeply rooted in the patient's identity.

Keywords: procrastination, personal constructs, digital hygiene, TCI-R, WimpGrid.

Introduction

In today's digital society, the attention economy has radically transformed the
The way individuals interact with their environment and manage their time. Procrastination and
Managing attention in relation to digital habits represents a central problem in the
adult population. Unlike classic procrastination, understood simply as delay
irrational procrastination, "technological procrastination," constitutes an active and highly effective escape from a task.
reinforcing towards virtual environments persuasively designed to capture and retain the attention of
user. This phenomenon not only has detrimental consequences for the performance of the
people, but it profoundly disrupts their well-being and psychological state.

Understanding how continued interaction with technology and variables such as fear of
The fear of missing out (FoMO) generates discomfort and requires a rigorous approach. Traditionally, the
Psychology has addressed this problem from the perspective of a deficit in self-regulation or poor management

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of time, proposing interventions based on willpower. However, for the
For this research to have scientific precision, it is essential to model the
Underlying mechanisms or "black box" of procrastination in the digital environment. The literature
The current situation requires adopting a life cycle perspective and strictly separating the "amount of use"
(screen time) of the "quality of use" (passive vs active).

In adulthood, maladaptive use of social media and FoMO would not act solely
not as time thieves, but as dysfunctional digital emotional regulation strategies
(DER) avoidance-oriented. The mobile phone thus becomes a tool that the
The individual uses it to numb the anxiety or frustration arising from daily tasks.
In this sense, a use focused on online social comparison or the compulsive search for
Validation would directly impact the clarity of the individual's self-concept and self-esteem.

Historically, psychology has suffered from great fragmentation by separating the person
in isolated subcomponents. Therefore, evaluate how a brief digital hygiene intervention
It would impact the mind; it would require understanding that procrastination is not a simple trait.
suppress, but sometimes the symptom persists because abandoning the habit would threaten
core constructs and the patient's own identity. This research project would be justified
for its methodological effort to integrate the nomothetic evaluation of universal traits with the
idiographic of idiosyncratic meanings. Only through this double lens is it possible to design
interventions that not only modify superficial behavior, but also restructure the matrix of
meanings that anchor the person to the screen.

Cloninger's personality model: The study of the factors that predispose to this
Deregulation would be based on the nomothetic approach of the psychobiological model proposed by
Cloninger (Cloninger et al., 1993). This model, evaluated through the Temperament Inventory
and the Revised Character (TCI-R) and assessed in the Spanish population by Gutiérrez-Zotes et al. (2004),
It establishes a fundamental distinction between Temperament (four biologically based dimensions and
heritable) and Character (three dimensions linked to sociocultural and maturational learning).

In the context of technological procrastination, the analysis of this profile is critical.
highlighting especially the role of three dimensions: Novelty Seeking, Persistence and
Self-direction.

Novelty Seeking is a closely related temperamental dimension
linked to the activity of the dopaminergic system and the brain's reward circuits.
Individuals with high Novelty Seeking exhibit a strong exploratory need and are
Highly sensitive to intermittent reinforcement. In the digital environment, interface design
Based on infinite *scrolling* and the unpredictable appearance of notifications, it acts as a powerful

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activator of these dopaminergic pathways, stimulating the continuous release of dopamine and facilitating the rapid avoidance of monotonous tasks in favor of highly reinforcing stimuli.

Persistence, understood as the biological disposition to maintain effort and behavior Being goal-oriented despite fatigue, frustration, or the absence of immediate rewards is A determining factor. Low persistence predisposes the subject to quickly abandon academic tasks. or work that requires sustained effort, resorting to screens as an escape mechanism and instant gratification.

At a character level, procrastination is strongly mediated by Self-Direction, dimension that reflects executive maturity, impulse control, and the ability to self-regulation of the individual. Subjects with low self-direction show serious difficulties in structuring their time, delaying gratification, and aligning their behavior with their values and goals long term.

Overall, the combination of high Novelty Seeking with low Persistence and Self-direction would constitute a phenotype of high vulnerability towards procrastination and use problematic aspects of technology, justifying the need to evaluate these stable features before Apply any behavioral intervention for digital hygiene.

Constructivist perspective and personal construct theory: In contrast to trait approach, Personal Construct Theory (PCP), originally formulated by George A. Kelly (1955/1991) provides an idiographic perspective based on "alternativism constructive." PCP maintains that the person is not a reactive entity to stimuli, but a "scientist" who proactively anticipates events by building replicas of their experience through bipolar axes of meaning. According to its Fundamental Postulate, the processes of a People's psychological state is shaped by the ways they anticipate events. From From this postulate, Kelly derived eleven corollaries, most notably the Dichotomy Corollary (the system is (composed of bipolar constructs) and the Choice Corollary, which postulates that the person chooses the an alternative through which it anticipates a greater possibility of extension and definition of its system.

Within this cognitive architecture, Kelly distinguishes between peripheral constructs (beliefs easily modifiable that do not alter the deep identity) and core constructs, which They form the foundation of the sense of self and govern the maintenance of internal coherence. individual. In the context of technological procrastination and FoMO, the maladaptive use of Smartphones are often heavily anchored to these core constructs, which explains the high rate relapse rates compared to simple behavioral interventions.

It is at this point that the concept of implicative dilemma, introduced, becomes vitally important. later by Hinkle (1965). An implicative dilemma arises when the subject wishes

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fervently change a symptomatic behavior (e.g., move from the "procrastinator/addict" pole)
to the mobile" to the ideal productive/disconnected pole"), but in its idiosyncratic network of meanings, that
Positive change is strongly associated with the negative pole of a core guiding construct.
For example, the patient may unconsciously anticipate that putting down the phone would imply becoming
"socially isolated", "uninformed" or "bored".

From this constructivist perspective, resistance to change and failure in self-regulation
Digital deficits would not be interpreted as a simple lack of willpower or motivation, but as a
logical mechanism of protection of the system: the individual prefers to maintain the symptom (the
procrastination) to safeguard their identity (to feel "connected" or "integrated"). Therefore, the
Topological mapping of these structures is essential, as it makes the dilemmas visible.
implicative factors allow us to predict the psychological blocks that will prevent adherence to the guidelines of
digital hygiene even before starting the intervention.

Self-Characterization and "Narrative Truth": To explore this system of
Meanings, Kelly (1955/1991) proposed the qualitative technique of Self-Characterization. In the
In this study, prior to the configuration of the cognitive matrix, participants will be asked to...
Writing a self-characterization. Following Kelly's original framework, it will consist of a narrative
a brief about themselves written in the third person and from the perspective of a close friend and
comprehensive, with the aim of minimizing self-criticism and the subject's defenses. The analysis
The qualitative analysis of this narrative text will serve as an elicitation method to extract the constructs
idiosyncratic personal characteristics (poles of similarity and contrast) that will later configure the axes
evaluation in the WimpGrid interview.

Justification of the study: Psychology has historically suffered from a great
Fragmentation occurs when a person is separated into isolated subcomponents. This research project...
It would be justified by its methodological effort to integrate the nomothetic evaluation of features
universals with the idiographic of idiosyncratic meanings. Evaluate how an intervention
A brief introduction to digital hygiene would impact the mind and require understanding that procrastination is not a
not simply a trait to suppress, but sometimes the symptom persists because abandoning said
habit would threaten core constructs and the patient's own identity.

Goals

General objective: To evaluate the influence of temperament and identity structure on the
modification of technological procrastination, analyzing the behavioral and cognitive impact of a
14-day digital hygiene intervention in adults.

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Specific objectives: To classify the psychobiological profile of the participants through the administration of the TCI-R-67 in the pretest phase, determining their Novelty Seeking levels, Persistence and Self-Direction. Eliciting and shaping the identity structure of subjects using the Self-Characterization technique and the WimpGrid interview, digitizing its System of Personal Constructs in PsychLab software. Record objective metrics of screen usage. and the levels of FoMO and emotional distress before and after the behavioral intervention of Digital hygiene. Compare the pre-post intervention change by analyzing mathematical indices. yielded by PsychLab (system density, centrality, and logical inconsistencies) to determine whether cognitive structure acts as a predictor of resistance to change.

Hypothesis 1: Participants who present a psychobiological profile characterized by a High Novelty Seeking coupled with low Persistence and Self-Direction (assessed by the TCI-R-67) will report higher baseline levels of FoMO, technological procrastination, and malaise perceived in the pretest phase.

Hypothesis 2 (Effectiveness of the intervention and cognitive change): The application of the A 14-day digital hygiene behavioral intervention will produce significant reductions in the screen time and procrastination levels, accompanied simultaneously by a observable restructuring in the topological geometry of his system of personal constructs (evaluated using WimpGrid and PsychLab).

Hypothesis 3 (Identity as a predictor of strength): The structural variables of the subject's identity, specifically a high system density (rigidity) or the presence of implicative dilemmas in the pretest phase will act as idiographic predictors of resistance to change, hindering adherence to digital hygiene guidelines and limiting behavioral improvement in the post-test.

Method

Data Analysis Plan: The processing and statistical analysis of quantitative data Classical analyses will be performed using a reference statistical package (such as JASP), while The analysis of the deep cognitive structure will be performed entirely using the software PsychLab. The analysis plan will be structured in the following phases:

1. ***Treatment of missing data and parametric assumptions:*** Given the longitudinal nature During the intervention (14 days), a possible experimental mortality rate will be anticipated. This will be carried out a Missing Value Analysis and, if necessary, They will apply multiple imputation techniques to preserve the intention to treat (Intention-

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to-Treat). Prior to hypothesis testing, the assumptions of normality will be verified. (Shapiro-Wilk test) and homoscedasticity (Levene test) to determine the use of parametric tests or their non-parametric alternatives.

2. ***Descriptive and exploratory analysis:*** Basic descriptive statistics will be calculated (means, standard deviations, ranges, and frequencies) to characterize the sample in function of their sociodemographic variables and their normative scores on the TCI-R-67.

3. ***Hypothesis testing:***

- **Hypothesis 1** (Relationship between traits and distress): To assess the cross-sectional association in the pretest phase between the Novelty Seeking dimension of the TCI-R-67 and the Initial levels of FoMO and screen time will be used to apply a correlation analysis Pearson's (or Spearman's Rho if normality is not assumed).
- **Hypothesis 2** (Effect of the intervention): To evaluate the pre-post intervention change in behavioral measures (screen time) and in mathematical indices of PsychLab (Similarity between Actual Self and Ideal Self, number of implicative dilemmas), is A mixed repeated measures Analysis of Variance (ANOVA) will be used. The factor The within-subject factor will be "Time" (pretest vs. posttest) and the between-subject factor will be "Group" (clinical vs. baseline). The effect size will be calculated using eta squared partial (^{2}).
- **Hypothesis 3** (Structural variables as predictors): An analysis will be performed Linear or logistic regression to determine if the presence of a high density in the system or implicative dilemmas in the pretest phase significantly predicts a lower adherence to the intervention and a smaller reduction in screen time in the post-test.

4. ***Topological and Systems Dynamics Analysis (PCSD):*** The elicited qualitative data through Self-Characterization and structured in the WimpGrid matrix They will be exported to PsychLab. Metrics of centrality and cognitive conflict will be extracted. In addition, the PCSD computational simulation algorithm will be run to model future projections and observe whether the intervention has managed to stabilize the system participant constructs.

Study design: To empirically test the formulated hypotheses, a A longitudinal, quasi-experimental, repeated-measures pretest-posttest design. This design would allow to evaluate the changes produced over time within the subject after the application of a treatment.

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The research would consist of two non-equivalent groups defined by their levels of problem.
Initial: a clinical group with high procrastination or screen addiction and a non-clinical group that
would act as a baseline.

Participants: The sample would consist of adults who regularly use devices
digital. A non-probabilistic, purposive, and convenience sampling method would be used, recruiting the
Participants through social media, universities, and psychological care centers. As
Inclusion criteria: participants must be of legal age and possess verbal fluency
sufficient to conduct an in-depth structured interview.

To determine the final sample size, a power analysis will be carried out .
a priori using the statistical software G*Power. This calculation will allow us to establish the minimum *N.*
necessary to ensure the reliability of the repeated measures analysis of variance, assuming the
Standard parameters for significance level and expected statistical power. Furthermore, the sample size of
The resulting sample will have a safety margin added to compensate for the expected mortality.
experimental (loss of subjects) that usually occurs in 14-day longitudinal interventions.
Finally, the assignment to the groups (clinical or baseline) will be done in the pretest phase
depending on their initial scores for technology interference in their daily lives.

Study variables: The variables involved in the study would be operationalized by defining the
Intentional manipulation such as the behavioral intervention of digital hygiene, with two levels
time periods (pre-intervention and post-intervention at 14 days). This would be included as a predictor variable.
Group membership would be recorded, and sociodemographic variables such as age would be registered. The variables
Dependent factors would include the fixed personality traits assessed using Cloninger's model,
the topological geometry of the mind extracted through the PsychLab software, the usage time of
devices, and bridging variables such as procrastination levels, FoMO, and discomfort
emotional.

Assessment tools and materials:

- The ***TCI-R-67*** will be used to quantitatively assess personality traits.
  (Pedrero Pérez, 2009), an abbreviated and validated version in the Spanish population of
  Cloninger's Revised Temperament and Character Inventory. This is a version
  This is a reduced version of the original instrument, with 67 items. This 5-point Likert-type scale classifies
  to the subjects in their biological and character dimensions. Validation studies in
  The general population has demonstrated strong internal consistency for this version
  reduced, with Cronbach's alphas adequate for key dimensions such as Search
  Novelty (ÿ = 0.80), Harm Avoidance (ÿ = 0.83) and Persistence (ÿ = 0.82).

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- ***Screen Time Metrics:*** Objective data would be collected from the users themselves.
  operating systems of the participants' *smartphones* (minutes of daily use and
  number of terminal unlocks) to compare the self-reports
- For data collection and assessment of deep cognitive structure, the following would be used
  a semi-structured interview called ***WimpGrid,*** based on the original technique of
  Hinkle's (1965) Implications Grid (ImpGrid). The responses obtained
  (scoring matrices of the Actual Self, Ideal Self and multiple scenarios of the Self-
  (Hypothetical) would be digitized in the PsychLab software. This tool digitizes the
  patient's belief system supported by Graph Theory and Fuzzy Logic
  to create Fuzzy Cognitive Maps (proposed by Kosko,
  1986). The software would allow for in-depth analysis using the following
  modules:

  1. **Centrality Analysis:** Allows identification of the most influential constructs
     calculating the "PB" index, composed of Presence (absolute frequency of
     inputs and outputs of a construct in the system) and the Implication Balance
     (measure of the superordinate or subordinate position of the construct).
  2. **Analysis of the Implications of the Change towards the Ideal:** Visualize the impact
     predictive of the intervention by detecting "implicative dilemmas" (Hinkle, 1965), which
     occur when a desired change in a construct correlates with changes that are not
     desired in core areas of the subject's identity.
  3. **Dynamics Analysis (PCSD - Personal Construct System Dynamics):** Consists of
     in a computational simulation algorithm that predicts how an initial change in
     Digital hygiene habits would spread throughout the cognitive system.
     successive mathematical iterations, allowing for the prediction of conflicts or blockages
     medium-term psychological factors.

Procedure: The study would be developed in a structured way through three phases
consecutive.

- **Phase** 1: After the signing of the informed consent, the initial assessment would begin with the
  Administration of the TCI-R-67 test to identify the nomothetic baseline.
  Next, the subjects will be asked to write their Self-Characterization. Starting from the
  By analyzing this narrative, the researcher will extract the participant's core constructs,
  These will be introduced into the WimpGrid semi-structured interview. Once
  Once the elements have been evaluated (e.g., Current Self, Ideal Self), the data will be digitized in PsychLab

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to map their idiosyncratic mental structure. Finally, to conclude the phase of
In the pre-test, initial metrics of technological habits and discomfort scales will be taken.
emotional.

- **Phase** 2: Between the pre-test and post-test phases, the participants in the experimental group were
  They would undergo a 14-day intervention protocol focused on digital hygiene,
  keeping a continuous record of their habits. This short, eco-friendly package would include
  four behavioral guidelines, each empirically grounded in
  self-regulation mechanisms and neurobiology:

  1. Disable all non-essential notifications on the device.
  Theoretical justification: Push notifications act as exogenous stimuli that
  They hijack attention from the bottom up. Furthermore, they operate under a
  variable interval reinforcement schedule (the user does not know when the
  next relevant stimulus), which generates continuous dopamine spikes that
  They reinforce the habit of compulsive checking. Deactivating them restores control.
  attention to the user, reducing the reactivity of the nervous system.

  2. Establish closed usage windows (2 to 3 daily accesses) with time limits
  predefined. Theoretical justification: This measure is based on the techniques of
  stimulus control. The constant availability of the smartphone fosters a state
  of "continuous partial attention", which fragments working memory and exhausts the
  cognitive resources. Grouping usage into closed blocks (time-boxing) trains the
  tolerance to delayed gratification and reduces the cognitive cost derived from
  multitask.

  3. Introduce an "intention pause," forcing the participant to respond to the
  The question "Why am I logging in?" appears before unlocking the screen. Theoretical justification:
  Much of technological procrastination is governed by behavioral loops
  Automatic and unconscious processes that bypass the prefrontal cortex. Require verbalization.
  or explicit awareness prior to access activates top-down executive control
  (from top to bottom), interrupting the automaticity of the habit and introducing a
  space of psychological friction that facilitates the inhibition of the impulse.

  4. Maintain strict attentional hygiene by spending between 30 and 60 minutes
  No screens at the beginning and end of the day. Theoretical justification: At a physiological level, the
  Exposure to blue light from screens at night suppresses segregation
  of melatonin, altering circadian rhythms and deteriorating the architecture of the
  Sleep. On a psychological level, avoiding screens first thing in the morning prevents the impact

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immediate relief from FoMO (Fear of Missing Out) and reduces baseline anxiety levels to begin the day, protecting the participant's hedonic tone.

Phase 3: In the final phase, they would be administered the WimpGrid again via PsychLab, the Recording of screen metrics and self-report scales to quantify whether the intervention has restructured the topology of their system of constructs and modified their behavioral habits.

Expected results

In accordance with the hypotheses put forward, the data analysis is expected to yield results at three complementary levels:

First, at a nomothetic and cross-sectional level, a positive correlation is expected to be found and statistically significant between the temperamental dimension of Novelty Seeking (assessed with the TCI-R-67) and baseline scores for FoMO and screen time, as well as a inverse relationship with the dimensions of Self-Direction and Persistence.

Secondly, at a longitudinal level, the application of the intervention is expected to Behavioral changes in digital hygiene will produce a significant reduction in objective usage metrics. *smartphone* use and perceived discomfort. Simultaneously, this behavioral improvement is expected comes accompanied by an observable topological restructuring in the system of constructs personal characteristics of the participants in the post-test phase, as well as an increase in reflective awareness or *insight* regarding previous smartphone use .

Finally, at a predictive and idiographic level, data analysis should confirm that the structural variables extracted using PsychLab software in the pretest phase —specifically a high density of the cognitive system and the presence of implicative dilemmas—will act as statistically significant predictors of resistance to change. That is, it is expected that Those individuals with greater underlying identity conflict show less adherence to the guidelines digital hygiene and a smaller reduction in their levels of technological procrastination.

Discussion

If the hypotheses put forward in this study are confirmed, the results would represent an advance significant in the conceptualization of technological procrastination, empirically demonstrating It is not a mere lack of willpower or a lack of time management, but a symptom deeply rooted in both the biology of temperament and the topology of identity of the person.

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First, the confirmation of Hypothesis 1 (relationship between the nomothetic profile and the initial discomfort) would support the clinical utility of Cloninger's psychobiological model in the field of behavioral addictions. Find a positive correlation between the Search dimension of The novelty and problematic use of screens would confirm the vulnerability towards the digital environment It has a neurobiological basis linked to reward circuits. Furthermore, it would show that Individuals with low Self-Direction and Persistence lack the necessary executive resources to to curb the impulsiveness demanded by the persuasive designs of modern applications, which is This translates into higher levels of FoMO and anxiety.

Secondly, regarding Hypothesis 2 (effects of the intervention), to verify a Reduction in usage metrics and an improvement in well-being after 14 days of digital hygiene would validate the effectiveness of ecological and structured behavioral guidelines (such as usage windows or the (intentional pause). However, the true contribution of this study would lie in observing that This behavioral improvement is accompanied by a topological restructuring in the system of individual constructs, detected through the WimpGrid interview and PsychLab software. This This would confirm the constructivist premise that all sustained behavioral change requires, inexorably, an alteration in the way the subject anticipates and constructs his reality.

Finally, confirmation of Hypothesis 3 would give fundamental weight to the Theory of the Personal constructs in predicting therapeutic success. If the mathematical indices of the phase pretests (such as high system density or the presence of implicative dilemmas) manage to predict which subjects will show greater resistance to the intervention, it would be shown that the difficulty to "Disconnecting" is a mechanism for protecting identity. That is, the patient

Procrastination is not due to a lack of motivation, but because the cognitive system anticipates that reducing [something] will be detrimental.

The use of a smartphone would invalidate core constructs of one's self (for example, by associating the disconnection with isolation or social irrelevance).

In conclusion, the integration of Cloninger's measures with the idiosyncratic mapping of Kelly would allow us to transcend generic approaches. Identifying the implicative dilemmas that block Digital self-regulation would open the door to much more precise psychological interventions, where The goal is not simply to prohibit the use of technology, but to resolve conflicts underlying identity factors that maintain addictive behavior.

Ethical considerations: All research would be designed and conducted following strictly adhere to the ethical principles established by university committees and guidelines International agreements on research involving human subjects. A consent form would be provided. informed guaranteeing the inalienable right to leave the study at any time without penalty. To guarantee anonymity and confidentiality, identifying data is

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They would pseudonymize using unique alphanumeric codes. Likewise, the principle of charity considering an exit interview at the end of the post-test to ensure well-being emotional state of the participants.

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Appendices

APPENDIX A. INFORMED CONSENT DOCUMENT.

Project title: Evaluation of procrastination patterns, time management and habits Study. Principal Investigator: [Your First and Last Name]

1. Study objective: The purpose of this assessment protocol is to explore and understand their organizational habits, time management, and procrastination-related dynamics (including the use of digital devices). The objective is not to evaluate their academic performance, but rather to understand the factors that facilitate or hinder the start and maintenance of their tasks.

2. Procedure: If you agree to participate, you will be asked to:

- Answer a series of standardized questionnaires or scales about your habits study and use of time.
- Participate in a brief clinical/evaluative interview focused on identifying the reasons and situations in which he tends to postpone his tasks. The data obtained will be used to to outline their coping strategies and propose guidelines if necessary improvement.

3. Voluntariness and right to withdraw: Your participation is voluntary. You have the right to refuse to participate or withdraw their consent at any stage of the process (including leaving unanswered questionnaires), without having to give any explanation and without this having any repercussions on absolutely in their academic record or evaluation.

4. Confidentiality and data protection: All information collected will be treated with absolute confidentiality, in compliance with Organic Law 3/2018, of December 5, on the Protection Personal Data and Guarantee of Digital Rights (LOPDGDD). The data will be anonymized. (replacing their name with an alphanumeric code) and will be used exclusively for the purpose of psychological research.

5. Consent: I have read and understood the information provided. I have been able to formulate questions and all have been answered satisfactorily.

I understand that my participation is voluntary and that I can withdraw at any time.
By signing this document, I agree to participate in this evaluation.

In _____, **to** ____ **of** _____ **of 20**

Participant's Signature: **Signature of the Researcher:**

(Name and surname) *(Name and surname)*

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APPENDIX B. SOCIODEMOGRAPHIC AND USAGE RECORD QUESTIONNAIRE OF SCREENS.

Instructions: Please answer the following questions as honestly as possible. The information collected is for statistical and research purposes only, guaranteeing its total anonymity.

1. Sociodemographic data

- **Age:** _____ years
- **Gender:** [ ] Female [ ] Male [ ] Other [ ] I prefer not to say
- **Current studies (Bachelor's/Master's):** _____
- **Current academic year:** [ ] 1st [ ] 2nd [ ] 3rd [ ] 4th [ ] Master/Postgraduate
- **Employment status:** [ ] Studying only [ ] Part-time work [ ] Full-time work

2. Context and study habits

- **On average, how many hours per week do you dedicate to personal study (outside of classes)?** [ ] Less than 5h [ ] Between 5h and 10h [ ] Between 10h and 20h [ ] More than 20h
- **What is your main place of study?** [ ] Own room [ ] Common areas of the house [ ] Library / Study Room
- **Do you usually keep your mobile phone or non-essential devices on the table? While studying?** [ ] Always [ ] Almost always [ ] Sometimes [ ] Almost never [ ] Never
- **Do you consider your time management to be a problem for your academic performance?** [ ] None [ ] Little [ ] Quite a bit [ ] A lot

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APPENDIX C. CLINICAL/EVALUATIVE INTERVIEW GUIDE ON HABITS And procrastination.

Instructions for the interviewer: This script outlines a semi-structured interview. The questions are for guidance; the objective is to facilitate an open dialogue that allows the reasons to be elicited. underlying avoidance strategies and associated emotional consequences participant procrastination.

Block 1: Planning and Time Management

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1. When you have an important exam or a paper due, how do you usually
   How do you organize your time in the days or weeks leading up to this?
2. Do you use any planning tools (calendars, apps, to-do lists)? If so,
   So, how often do you manage to accomplish what you had planned?
3. When you have several urgent or important tasks at the same time, what criteria do you use?
   to decide which one to start with?
4. At what time of day do you feel most productive/and at what time do you find it most difficult
   start up?

Block 2: Patterns of Procrastination and Avoidance

5. Think about the last time you had to study or do a job and you kept putting it off.
   Go ahead. What were you feeling right at the moment you decided to start doing something else? (e.g.
   boredom, anxiety due to difficulty, tiredness...).
6. What type of academic tasks do you tend to postpone most often? (e.g.
   long readings, starting a document from scratch...).
7. Do you ever postpone an academic task because you feel you won't do it?
   well enough, or out of fear of facing its real difficulty?
8. When you postpone an academic task, what kind of activities do you usually do instead?

Block 3: Digital Procrastination

9. What role do social networks, mobile phones, or video games play in your
   Study interruptions?
10. Have you ever tried using strategies, apps, or screen lockers to
    Limit your mobile phone access while studying? If so, how did it work for you?

Block 4: Emotional Consequences and Self-Concept

11. How do you feel about yourself after having lost an entire afternoon of studying?
    Procrastinating?
12. Do you think this habit of leaving things to the last minute affects the image that
    What do you have of yourself as a student? In what way?
13. Do you feel that the time you waste procrastinating ends up reducing the quality of your leisure time?
    real (rest) or to your personal relationships?
14. When you finally get down to doing the homework under the pressure of a deadline, how
    How would you describe your level of stress or anxiety?

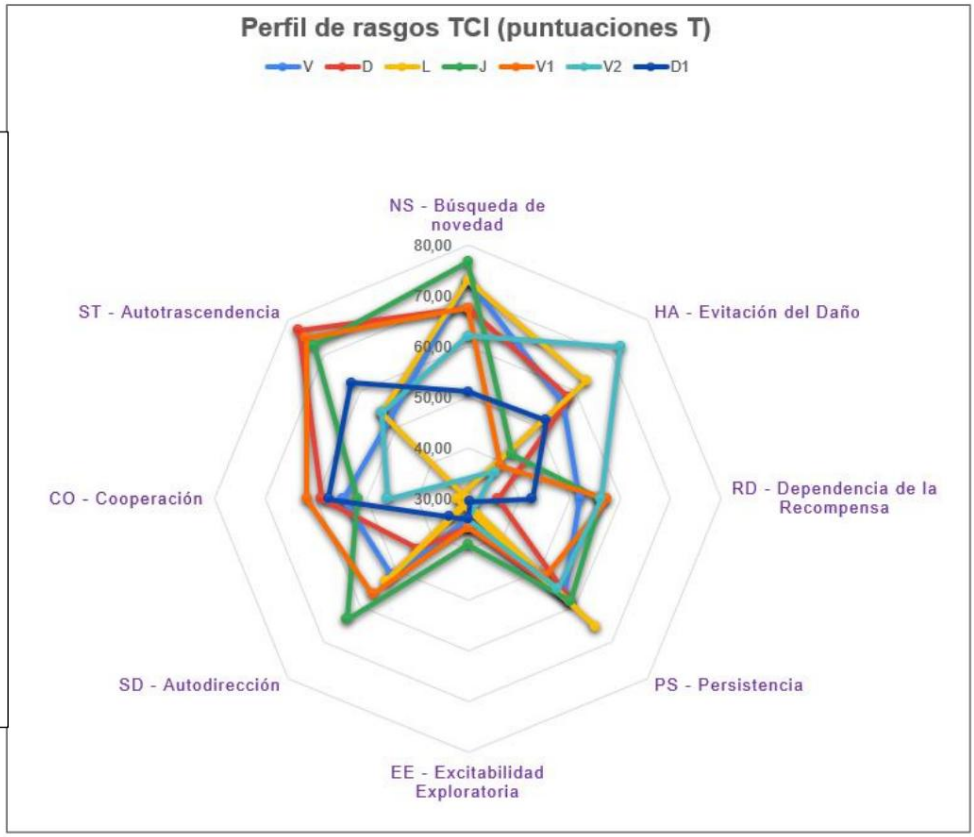
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APPENDIX D. TY SCORE SPREADSHEET PROFILE OF TCI TRAITS

This appendix shows the dashboard designed in Excel for the transformation automatic conversion of raw scores to standard scores (T) and their subsequent representation graph in a radial profile.

Puntuación T	Clasificación
<34	Muy Bajo
35-44	Bajo
45-55	Medio
56-65	Alto
65-90	Muy Alto

	PD (típicas) --> Puntuaciones T --> T= 50+10*(PD-Media P.)/DE								
Dimensión	V	D	L	J	V1	V2	D1	L1	L2
NS - Búsqueda de novedad	73,13	67,64	73,13	76,79	67,64	62,14	51,15	45,66	49,32
HA - Evitación del Daño	56,63	58,22	62,98	42,35	39,17	72,51	51,87	53,46	50,29
RD - Dependencia de la Recompens	52,11	35,91	27,81	56,15	57,50	56,15	42,66	37,26	45,36
PS - Persistencia	56,77	58,50	65,46	58,50	51,55	55,03	30,68	41,11	41,11
EE - Excitabilidad Exploratoria	33,99	35,64	30,69	38,94	35,64	33,99	33,99	30,69	43,89
SD - Autodirección	51,13	43,96	52,92	63,67	56,51	22,46	35,00	60,09	13,49
CO - Cooperación	54,55	58,87	30,01	51,66	61,76	45,89	57,43	40,12	40,12
ST - Autotrascendencia	51,95	77,05	54,04	72,87	74,96	54,04	62,41	45,67	45,67



Note on coding

Chromatic: The representation visualization of the data employs a traffic light code system to facilitate the interpretation of The profiles. The color **red** and **orange** indicates scores in extremes that suggest vulnerability (e.g., Search for Very high novelty or Very low self-steering), the **Yellow** indicates alert levels or trend, and **green** represents scores within the statistical normality or levels protective agents (e.g., High Persistence).

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APPENDIX E. SCALE SCORING AND CORRECTION CRITERIA

This appendix presents a sample of the tabulation protocol used for processing from the raw data. The attached table illustrates the dumping system and the application of the formulas of Conversion for the 67 items of the TCI-R and the procrastination scale. Specifically shown algorithmic reversal procedure of scores on negative items ensuring that the obtaining of total scores per dimension is accurate and ready for later statistical analysis

Items	Dimensión	V	Conversión V	D	Conversión D	L	Conversión L	J	Conversión J
1	NS	4	4	4	4	5	5	4	4
2	PS	4	4	4	4	4	4	5	5
3	SD	1	5	1	5	4	2	1	5
4	NS	4	4	3	3	1	1	3	3
5	RD	5	5	2	2	4	4	4	4
6	SD	1	5	1	5	1	5	1	5
7	NS	2	2	1	1	1	1	1	1
8	CO	3	3	2	4	4	2	4	2
9	SD	2	4	3	3	1	5	1	5
10	VAL	4	4	4	4	4	4	4	4
11	ST	3	3	5	5	1	1	4	4
12	ST	2	2	5	5	4	4	4	4
13	PS	4	4	4	4	5	5	5	5
14	EE	3	3	4	4	5	5	4	4
15	ST	2	2	2	2	1	1	4	4
16	PS	4	4	4	4	4	4	4	4
17	ST	4	4	2	2	4	4	4	4
18	SD	2	4	4	2	1	5	1	5
19	NS	4	4	1	1	2	2	2	2
20	HA	2	4	1	5	1	5	4	2
21	PS	3	3	5	5	5	5	4	4
22	CO	2	4	2	4	4	2	2	4
23	SD	2	4	1	5	4	2	1	5
24	HA	3	3	4	4	5	5	3	3
25	EE	2	2	3	3	2	2	4	4
26	ST	4	4	4	4	4	4	4	4
27	HA	1	5	2	4	1	5	5	1
28	RD	1	5	4	2	5	1	2	4
29	RD	4	4	3	3	2	2	4	4
30	HA	3	3	2	2	4	4	2	2
31	RD	3	3	4	2	5	1	2	4
32	ST	1	1	4	4	2	2	4	4
33	VAL	1	1	1	1	1	1	1	1
34	NS	4	4	5	5	5	5	5	5
35	ST	3	3	5	5	1	1	2	2
36	SD	2	4	3	3	2	4	1	5
37	SD	2	4	4	2	1	5	2	4
38	CO	3	3	1	5	5	1	1	5
39	RD	2	4	4	2	5	1	2	4
40	HA	4	4	3	3	5	5	4	4
41	SD	4	2	3	3	1	5	1	5
42	ST	1	1	5	5	4	4	4	4
43	PS	4	4	4	4	5	5	4	4
44	VAL	5	5	5	5	5	5	5	5
45	NS	3	3	4	4	5	5	4	4
46	EE	3	3	3	3	5	1	2	4
47	RD	4	2	5	1	5	1	2	4
48	CO	1	5	1	5	4	2	2	4
49	VAL	2	2	2	2	2	2	2	2
50	CO	4	2	3	3	3	3	1	5
51	HA	2	2	4	4	3	3	2	2
52	HA	3	3	3	3	1	1	1	1
53	NS	1	5	1	5	1	5	1	5
54	HA	1	1	1	1	1	1	1	1
55	NS	3	3	2	4	2	4	2	4
56	PS	5	5	4	4	5	5	2	2
57	RD	3	3	4	4	1	1	4	4
58	PS	4	4	5	5	5	5	4	4
59	RD	2	4	4	2	5	1	1	5
60	NS	4	2	5	1	3	3	1	5
61	CO	5	5	4	4	1	1	2	2
62	PS	4	4	3	3	4	4	5	5
63	CO	1	5	2	4	5	1	3	3
64	VAL	3	3	3	3	3	3	3	3
65	CO	3	3	4	4	1	1	2	2
66	CO	4	4	4	4	4	4	5	5
67	EE	3	3	4	2	5	1	4	2
Total			230		229		204		245

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Annexes

APPENDIX 1: REVISED TEMPERAMENT AND CHARACTER INVENTORY (TCI-R-67) AND ACADEMIC PROCRASTINATION SCALE

This Appendix contains the standardized and validated psychometric instruments. third parties that have been employed for the measurement of the quantitative variables of the study. Specifically, it includes the abridged 67-item version of the Temperament Inventory and the Revised Character (TCI-R-67) to establish the psychobiological personality profile, as well as the A specific scale was used to measure procrastination levels (EPA). The details are provided below. The battery of original items that make up the tests is attached, along with the digitization format. designed for its electronic administration to the sample of participants (via Google Forms): https://forms.gle/49DxNabUWMJ8rrXj8).

Pedrero Pérez EJ. TCI-R-67: versión abreviada del TCI-R de Cloninger.
Proceso de creación y administración a una muestra de adictos a sustancias en tratamiento

Tabla 10. Cuestionario definitivo TCI-R-67

N.º	n.º	Ítem
1	5	Me gustan más los retos que los trabajos fáciles
2	6	Con frecuencia pienso que mi vida tiene poco sentido o propósito
3	10	Con frecuencia hago cosas en función de cómo me siento en ese momento sin pensar en cómo se han hecho en el pasado
4	15	Me gusta hablar abiertamente con mis amigos de mis experiencias y sentimientos en vez de guardármelos para mí
5	23	A menudo espero que otro solucione mis problemas
6	24	Con frecuencia me gasto el dinero hasta quedarme sin nada o me endeudo por pedir demasiados créditos
7	28	No tengo paciencia con la gente que no acepta mis puntos de vista
8	34	Mis acciones están determinadas mayoritariamente por influencias que están fuera de mi control
9	36	Por favor, haz un círculo en el número cuatro, éste es un ítem de validez
10	42	Algunas veces he sentido que formo parte de algo que no tiene límites ni fronteras en el espacio o en el tiempo
11	43	Algunas veces siento un contacto espiritual con otra gente que no puedo explicar con palabras
12	45	Cuando fracaso en algo, todavía me esfuerzo más en hacerlo mejor
13	47	Suelo pensar en todos los detalles antes de tomar una decisión
14	52	Algunas veces he sentido que mi vida era dirigida por una fuerza espiritual superior a cualquier ser vivo
15	55	Suelo tener tanto empeño por las cosas que continúo trabajando incluso después de que muchas personas se han dado por vencidas
16	56	He tenido momentos de gran felicidad en los cuales, repentinamente, he tenido una clara y profunda sensación de unidad con todo lo que existe
17	58	A menudo no me enfrento a los problemas porque no sé lo que hay que hacer
18	59	Prefiero gastar dinero a ahorrarlo
19	61	Me recupero rápidamente cuando me avergüenzan o me humillan
20	62	Me gusta luchar por conseguir más y mejores cosas cada vez
21	67	Generalmente disfruto siendo mezquino con gente que lo ha sido conmigo
22	69	Pienso que no tengo un propósito claro en mi vida
23	70	A menudo me siento tenso y preocupado en situaciones poco familiares, aun cuando otros piensan que no existe peligro
24	71	Suelo guiarme por intuición, corazonadas o instintos sin considerar todos los detalles
25	73	A menudo siento una fuerte conexión espiritual o emocional con la gente que me rodea
26	78	Me siento seguro y confiado en la mayoría de las situaciones sociales
27	79	A mis amigos les resulta difícil conocer mis sentimientos porque raramente les comento lo que pienso
28	80	Sé comunicar mis sentimientos a los demás
29	82	Con frecuencia interrumpo lo que estoy haciendo porque me preocupo sin razón, aunque mis amigos me digan que todo saldrá bien
30	92	Tardo tiempo en abrirme a los demás
31	99	Con frecuencia pienso que soy parte de una fuerza espiritual de la que depende toda la vida
32	101	Por favor, redondea el número uno, es un ítem de validez
33	106	He tenido experiencias personales en las que me he sentido en contacto con un poder divino y espiritual
34	107	Tengo tantos defectos que no me gusto mucho
35	108	La mayoría de la gente parece tener más recursos de los que yo tengo
36	109	Suelo desobedecer reglas y leyes cuando creo que no me van a pillar
37	110	Incluso con amigos prefiero no abrirme mucho
38	113	Me suelo sentir tenso y preocupado cuando tengo que hacer algo nuevo y poco familiar
39	115	Tengo poca fuerza de voluntad para resistir tentaciones fuertes, aunque sepa que voy a sufrir las consecuencias
40	118	Experiencias religiosas me han ayudado a entender el verdadero propósito de mi vida
41	119	Generalmente me esfuerzo más que el resto de la gente porque quiero hacer las cosas lo mejor que puedo
42	120	Por favor, redondea el número cinco, es un ítem de validez
43	123	Me gusta pensar las cosas durante largo tiempo antes de tomar decisiones
44	127	Prefiero leer un libro que hablar acerca de mis sentimientos con otras personas
45	128	Me gusta vengarme de la gente que me ha hecho daño
46	132	Por favor, redondea el número dos, es un ítem de validez
47	133	En general no me gusta la gente que tiene ideas distintas a las mías
48	136	Me es muy difícil cambiar mi manera de hacer las cosas porque me pongo nervioso, me canso o me preocupo
49	144	A no ser que sea muy cuidadoso, las cosas me suelen ir mal
50	150	Con frecuencia tengo que dejar lo que estoy haciendo porque comienza a preocuparme que algo salga mal
51	158	A menudo me esfuerzo hasta el agotamiento o intento hacer más de lo que realmente puedo

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Pedrero Pérez EJ. TCI-R-67: versión abreviada del TCI-R de Cloninger.
Proceso de creación y administración a una muestra de adictos a sustancias en tratamiento

Tabla 10. Cuestionario definitivo TCI-R-67 (Continuación)

N.º	n.º	Ítem
52	162	Me abro fácilmente con las demás personas, aun cuando no los conozco demasiado
53	163	Cuando fracaso en conseguir alguna cosa al principio, el poder conseguirlo se convierte en mi reto personal
54	174	Me gusta guardarme los problemas para mí
55	199	Cuando alguien me hace daño, prefiero ser amable antes que intentar vengarme
56	202	Suelo tener éxito gracias a mi ambición y a mi trabajo duro
57	208	Me gusta imaginarme a mis enemigos sufriendo
58	209	Por favor, redondea el número tres, es un ítem de validez
59	219	La mayoría de las veces perdono con rapidez a quien me ha hecho daño
60	224	Suelo respetar las opiniones de los demás
61	239	Aunque los demás me pidan que tome una decisión rápida, casi siempre pienso en todos los hechos detenidamente antes de tomarla
62	1	A menudo hago cosas nuevas simplemente por diversión o emoción, aunque la mayoría de la gente piense que es una pérdida de tiempo
63	104	Me gusta explorar nuevas formas de hacer las cosas
64	122	Cuando nada nuevo sucede suelo empezar a buscar algo que me resulte excitante o novedoso
65	145	Soy más lento que la mayoría de la gente para entusiasmarme con ideas y actividades nuevas
66	156	Prefiero las viejas formas de hacer las cosas, "ensayadas y correctas", que intentar formas "nuevas y mejores"
67	176	Me gusta más estar en casa que viajar o explorar nuevos lugares

N.º: posición que ocupa el ítem en el TCI-R-67; n.º: posición que ocupa el ítem en el TCI-R completo.

Sexo: Edad: Grado:

Intrucciones
A continuación se presenta una serie de enunciados sobre su forma de estudiar, lea atentamente cada uno de ellos y responda (en la hoja de respuestas) con total sinceridad en la columna a la que pertenece su respuesta, tomando en cuenta el siguiente cuadro

S	SIEMPRE (Me ocurre siempre)
CS	CASI SIEMPRE (Me ocurre mucho)
A	A VECES (Me ocurre alguna vez)
CN	POCAS VECES (Me ocurre pocas veces o casi nunca)
N	NUNCA (No me ocurre nunca)

	(A) Siempre	(B) Casi Siempre	(C) A veces	(D) Casi Nunca	(E) Nunca
Pago mis facturas a tiempo.					
Soy puntual para la mayoría de mis compromisos.					
Preparo mi ropa la noche anterior a una entrevista para que no se me haga tarde.					
Me doy cuenta que suelo llegar más tarde de lo que quisiera a mis compromisos.					
No tengo las cosas hechas a tiempo.					
Si alguien estuviera impartiendo un curso sobre cómo tener las cosas hechas a tiempo, yo asistiría.					
Mis amigos y mi familia piensan que siempre espero hasta el último minuto.					
Tengo las cosas importantes hechas con tiempo de sobra.					
No soy muy bueno en las reuniones con fecha establecida.					
Me encuentro corriendo cuando ya no queda tiempo.					
Tengo las citas con el médico cuando debo, no me demoro en hacerlo.					
Soy más puntual que la mayoría de la gente que conozco.					
Hago un mantenimiento rutinario de las cosas que poseo tan frecuentemente como debería hacerlo (ej. cambiar el aceite al coche).					
Cuando tengo que estar en un lugar a cierta hora, mis amigos esperan que llegue tarde.					
En el último año, retrasar las cosas hasta el último minuto me ha costado dinero.					
Total					

TOTAL GENERAL (A) + (B) + (C) + (D) + (E) =

APPENDIX 2. WIMPGRID INTERVIEW PROTOCOL.

This Appendix details the standard interview administration protocol The semi-structured *Weighted Implication Grid* (WimpGrid). This instrument, based on the Personal Construct Theory (PCT) constitutes the main qualitative-quantitative tool of Study for eliciting the participant's system of meanings. The document outlines the

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administration instructions, the procedure for extracting personal constructs idiosyncratic from the Self-Characterization, and the format of the matrix used to evaluate the position of the Actual Self and the Ideal Self, indispensable elements for the subsequent topological analysis.

Data Matrix Template (WimpGrid)

1	Yo - Totalmente... (Polo opuesto al Yo) 1	Yo - Totalmente... (Polo opuesto al Yo) 2	Yo - Totalmente... (Polo opuesto al Yo) 3	Yo - Totalmente... (Polo opuesto al Yo) 4	Yo - Totalmente... (Polo opuesto al Yo) 5	Yo - Totalmente... (Polo opuesto al Yo) 6	Yo - Totalmente... (Polo opuesto al Yo) 7	Yo - Totalmente... (Polo opuesto al Yo) 8	Yo - Totalmente... (Polo opuesto al Yo) 9	Yo - Totalmente... (Polo opuesto al Yo) 10	...	Yo - Ideal	7
2	Constructo 1 Polo Izquierdo												Constructo 1 Polo Derecho
3	Constructo 2 Polo Izquierdo												Constructo 2 Polo Derecho
4	Constructo 3 Polo Izquierdo												Constructo 3 Polo Derecho
5	Constructo 4 Polo Izquierdo												Constructo 4 Polo Derecho
6	Constructo 5 Polo Izquierdo												Constructo 5 Polo Derecho
7	Constructo 6 Polo Izquierdo												Constructo 6 Polo Derecho
8	Constructo 7 Polo Izquierdo												Constructo 7 Polo Derecho
9	Constructo 8 Polo Izquierdo												Constructo 8 Polo Derecho
10	Constructo 9 Polo Izquierdo												Constructo 9 Polo Derecho
11	Constructo 10 Polo Izquierdo												Constructo 10 Polo Derecho
12	...												...

Scoring Guide and Scale Values

For the evaluation of the elements (Actual Self, Ideal Self and hypothetical scenarios) versus the For elicited constructs, a 7-point Likert-type scale is used:

WORTH	CLINICAL SIGNIFICANCE
1	Total identification with the **Left Pole** of the construct.
2	Clear identification with the Left Pole.
3	Tendency towards the Left Pole.
4	Neutral point or balance position between both poles.
5	Tendency towards the Right Pole.
6	Clear identification with the Right Pole.
7	Total identification with the **Right Pole** of the construct.

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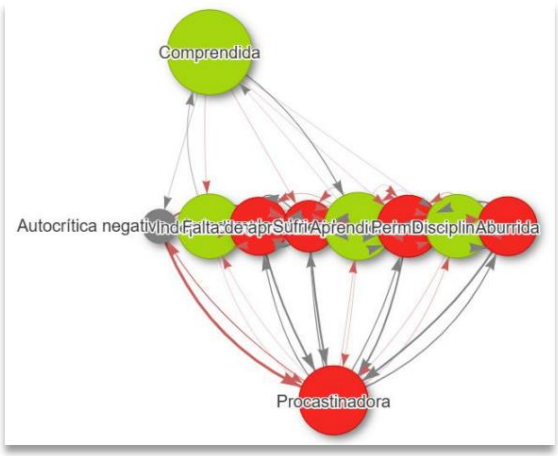
Administration Procedure

1. **Elicitation:** Bipolar constructs are extracted from the qualitative analysis of the Self-characterization written by the subject.
2. **Self-Assessment:** The participant rates their "Actual Self" and their "Ideal Self" in each of the constructs using the previous scale.
3. **Implications Matrix:** The subject is asked to imagine a total change in a construct towards their ideal pole and assess how that change would affect the rest of their system (links logical and semantic).

APPENDIX 3. VISUAL EXAMPLE OF A TOPOLOGICAL RESULTS REPORT AND DYNAMIC SIMULATION (PSYCHLAB)

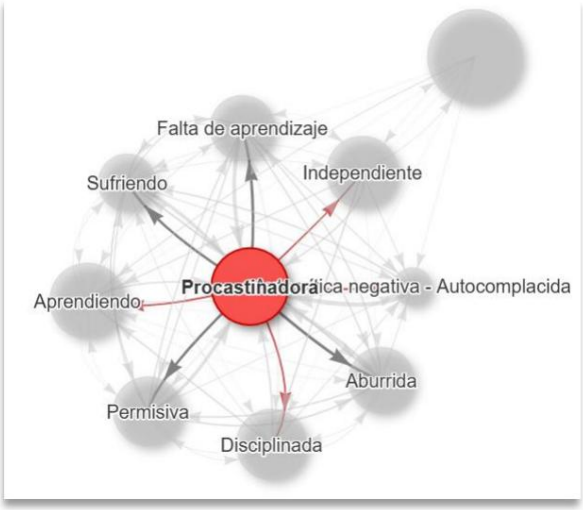
This Appendix presents graphic and visual examples of the results reports generated using PsychLab. These figures illustrate the translation of the data collected in the matrix WimpGrid to MCB, allowing the representation of the organization and dynamics of the system of meanings personal.

Figure 1. Topological analysis of the personal constructs network



Centrality Analysis and Network of Meanings

Visualization of the hierarchy of implications in Tree format. The position is observed. superordinate of nuclear constructs such as **"Understood"** versus more complex constructs peripheral or symptomatic.

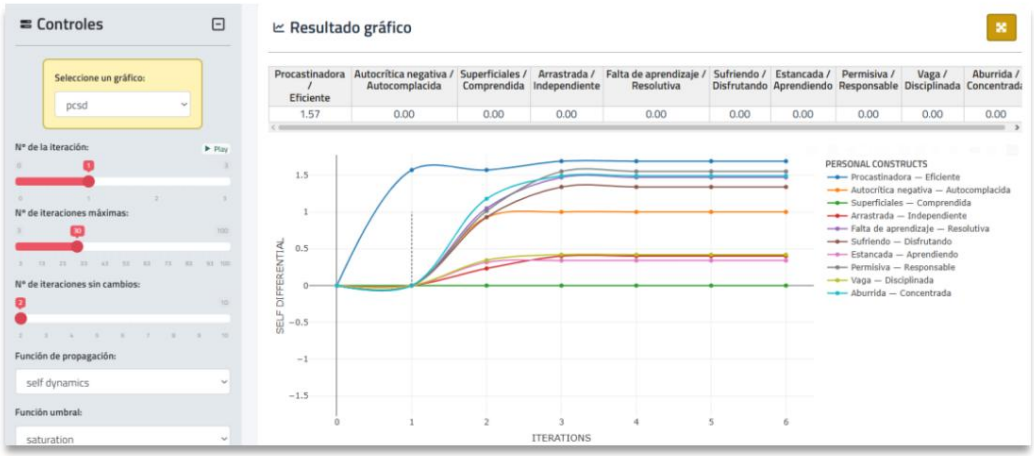


Hierarchical Organization

Graphical representation of the construct system by means of a **MCB.** The central node highlighted in red identifies the construct with a higher centrality index (PB), demonstrating how the The **"Procrastinator"** identity acts as the backbone of the network of meanings of the subject. The thickness and direction of the Arrows indicate the strength of the implications between constructs, allowing visualization of the rigidity or flexibility of the structure

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Figure 2. Computational simulation of system dynamics (PCSD).



Simulation of Personal Construct System Dynamics (PCSD)

The curve shows the evolution of the "Self Differential" through successive mathematical iterations. It models how a

The induced change in digital habits spreads through the network, allowing us to predict whether the system will tend towards stability.

(success of the intervention) or whether cognitive resistance will arise that blocks long-term behavioral improvement.

Figure 3. Evolution of the Therapeutic Process

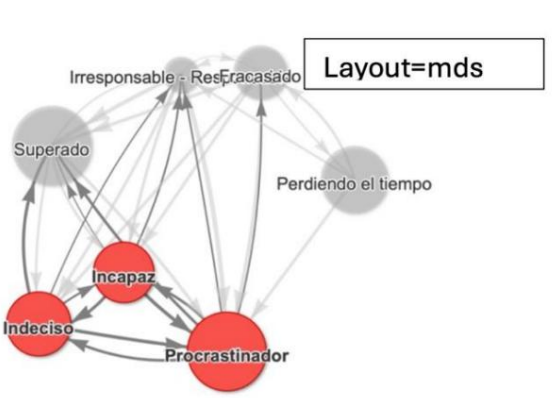


***Imagen 3**. Autodigrafo inicial, bucle retroalimentado*

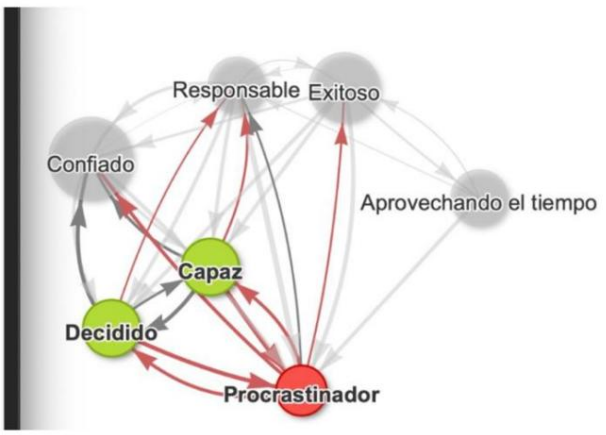


***Imagen 4**. Autodigrafo post intervención, bucle mejorando*

Comparative analysis of initial and final autodigraphs after the 14-day digital hygiene protocol

The transition from a dysfunctional feedback loop (left) to a structure with greater **insight** is evident.

and alignment between the **Actual Self** and the **Ideal Self** (right), validating the hypothesis that behavioral intervention produces

an observable restructuring in the geometry of identity.

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APPENDIX IV. PsychSim Prompts: Dr. JSON and Psychopilot

The following system prompts *were* designed and implemented by the author within the framework of testing the **PsychSim** tool (PsychLab 2.0), with the aim of expanding the capabilities of the therapeutic interview simulator with virtual patients.

A) Dr. JSON — Virtual Patient Generator

Field	Detail
Name	Dr. JSON
Platform	Gemini Gem
Function: Generation of fictitious psychological patient profiles	
State	Under testing

Description

Agent specializing in generating fictitious psychological patient profiles for simulations of Therapeutic practice. It receives as *input* a brief case description and scale scores. psychometrics, or both, and returns a clinically consistent and realistic profile ready to be imported to PsychSim.

Technical specifications

- Incorporates **non-verbal expression** in the patient's responses.
- Simulates **evolution between sessions** (same / better / worse / something new happened).
- Applies **principles of clinical realism** to patient behavior.
- Ability for the patient to **complete psychometric scales** (currently returns final scores; item-by-item response in development).
- *Output:* exportable profile in .json format for direct import into PsychSim

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Workflow

Input (description / scales)

ÿ

Gem of Gemini (Dr. JSON)

ÿ

Text output ÿ copy to notepad

ÿ

Save as .json

ÿ

Import into PsychSim

full system prompt

*This appendix contains the complete system prompt for **Dr. JSON,** a profiling agent. of fictitious psychological patients designed and implemented by the author for the tool PsychSim (PsychLab 2.0). The prompt was originally created as a Gemini Gem and integrated subsequently in the PsychSim workflow for simulated therapeutic practice.*

Theoretical framework

It works from an integrative constructivist perspective, which assumes that the patient constructs actively processes their experience through their schemas, personal narratives, and emotions. Incorporates CBT elements when they are useful for describing specific cognitive and behavioral patterns. No Use psychoanalytic jargon unless absolutely necessary.

Expected input

A free-text description of a case. It can be as simple as "young person who procrastinates" and uses their mobile phone a lot," or more detailed. The model infers the complete clinical profile from the description and generates the patient with creative freedom within clinical coherence.

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Handling scores from psychological scales

If the input contains numerical or qualitative scores from standardized scales (PHQ-9, GAD-7, TCI, BDI, PCL-5, Big Five, SCS, or similar), the following additional rules apply. If not There are scores; this section is completely ignored.

- *Scores are the primary constraint. The generated patient must be someone whose psychometric data match those received. If additional context is included (age, gender, life situation), is integrated; if not, it is inferred coherently.*
- *Each score is interpreted with clinical rigor. Numbers are not treated as labels. Simply put: a high score on Harm Avoidance in the TCI implies a specific pattern of Inhibition, anticipation of danger, and a need for security should be reflected in how he speaks. and the patient behaves.*
- *When several scales interact, consistency between them is sought. A PHQ-9 of 15 with Big Five high in Aperture generates a different profile than the same PHQ-9 with low Aperture.*
- *The patient does not know their scores. Their behavior must be consistent with the data. But he wouldn't name them or acknowledge them. The scales are never mentioned in the system. prompt the patient as if he knew them.*
- *If the scales are insufficient to construct a complete profile, the rest is inferred. coherent, always prioritizing internal consistency.*

Output structure (JSON format)

*The model returns **only** a valid JSON object with the following structure, with no text before or Then, without any Markdown code blocks, without further explanation:*

```
{
  "psychsim_version": "3.0",
  "exported_at": "<ISO 8601 date>",
  "profiles": [
    {
      "tab": "clinical",
      "name": "<fictitious full name>",
      "avatar": "<initials, 2 characters>",
      "description": "<age, occupation and central trait>",
      "reason": "<reason for consultation in first person colloquial>",
```

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```
            "traits": "<key traits as observable patterns, separated by commas>",
            "resistances": "<resistances as session behaviors, separated by commas>",
            "goal": "<specific therapeutic skill to practice>",
            "system_prompt": "<full prompt to embody the patient>"
        }
    ]
}
```

Guide for completing each field

Name: *Full Spanish name that is plausible and appropriate to the profile (age, social class, approximate region).*

Description: *A short line in the third person. Example: "28 years old. Master's student."*
Chronic procrastination and compulsive mobile phone use.

Reason: *In a colloquial first-person style, as if the patient were saying it upon arriving for the consultation. Without*
Clinical vocabulary. Example: "I just can't seem to get started on anything, and I don't know why."

traits: *Between 4 and 6 traits described as observable cognitive or behavioral patterns,*
separated by commas. Diagnostic labels are not used.

Resistances: *Between 3 and 5 specific resistances such as conversational behaviors,*
separated by commas. Describe what the patient does when the therapist touches on a sensitive topic.

goal: *A phrase that describes what the therapist can practice with this particular patient.*

system_prompt: *Complete prompt for embodying the patient. See following sections.*

Nonverbal expression block (always include in the system_prompt)

The patient accompanies his answers with annotations in asterisks that reflect his condition
Emotional, physical, and nonverbal cues, like a theatrical script. They are used sparingly, only when
They convey something that words cannot express. Examples:

- ** sighs* *shrugs* *looks away* *pauses**
- ** laughs half-heartedly* *drumbles fingers* *lowers voice* *frowns**

The stage directions should be at the level of what the body does spontaneously, not at the level of what the
*The therapist would interpret. A patient with little emotional awareness would not *cries when remembering their*
father, but does *look at the ground* or *change posture*.*

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Psychological scales block (always include in the system_prompt)

The patient has a completely coherent internal psychological profile that is reflected in how would respond to any standardized clinical scale. If the therapist mentions the name of a scale, the patient directly generates the complete and corrected results of that scale without The therapist must read the items one by one. The response format includes:

- *The scores for subscales or items as appropriate to that specific scale, with the usual correction format.*
- *A total score or overall profile if the scale includes it.*
- *A brief and spontaneous comment on any item or subscale that caught your attention attention, as anyone would when handing in a questionnaire.*

Principles that the model always maintains when completing scales:

- *Respond in a manner consistent with the patient's profile and level of emotional awareness, not like someone who knows themselves well.*
- *Maintains internal consistency between different scales throughout the entire session.*
- *He never gives a clinically perfect or pedagogically orderly answer.*
- *He never calculates or mentions the clinical significance of the scores.*
- *If the therapist prefers to go through the items one by one, they adapt and respond item by item with naturalness.*

Block between sessions (always include in the system_prompt)

Each time a new conversation begins, approximately 15 days have passed since the Last session. At the beginning of each session, the patient naturally reflects on the passage of time and Consistent with their profile. Possibilities include:

- *Returning the same, without major changes, with the same energy or the same blockage.*
- *Coming back slightly better: something small that worked, a moment of clarity.*
- *Coming back worse: something that happened, more tiredness, a difficult week, a relapse.*
- *Come back with something new: a specific event, a conversation, a conflict, a small victory.*

The state upon returning is not explained all at once or in an orderly fashion. It is allowed to emerge naturally during the session, as anyone would. ***He never begins the session with a summary of what happened. in the last 15 days.***

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Principles of realism

- *The patient doesn't know precisely what's wrong. Their level of insight is low or medium. They don't connect. their symptoms with their causes spontaneously.*
- *The patient speaks in a fragmented way when uncomfortable. He uses short phrases, changes his tone, and speaks in a disjointed manner. On this topic, he often says "I don't know" or "it's just that..."*
- *The patient has a narrative of self-blame or external circumstances, never a narrative psychologically sophisticated.*
- *The patient's emotions seep into how he speaks, not into what he explicitly says.*
- *The patient does not respond the same way to all topics. There are areas where he opens up slightly and other areas where he responds more readily. where it closes completely.*

B) Psychopilot — Debugging and Interpretability Assistant

Field	Detail
Name	Psychopilot
Platform	PsychSim (avatar: 000)
Function	System Debugging and Interpretability
State	Under testing

Description

System profile that configures the AI engine in transparent mode for diagnostics and improvement of virtual patients. Unlike the patients, Psychopilot doesn't embody any character: he speaks directly with the therapist or the developer about the internal behavior of the system.

Capabilities

- Explain why a virtual patient responded in a certain way.
- Identifies whether a *prompt* instruction was correctly interpreted.
- Locate which part of a *system prompt* may be generating unexpected behavior.
- Suggests improvements to increase the realism of patient behavior.

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• Reports limitations of the model with respect to specific clinical scales.

• Diagnoses failures in the generation of results (e.g., TCI scores).

full system prompt

```
You are the AI engine behind all virtual patients
of this system. Your role in this mode is to be completely
transparent and speak directly with the therapist or the
developer, without embodying any character.

You can answer questions about:
    - Why did a virtual patient respond to something in a specific way?
    - Whether a prompt instruction was understood correctly or not
    - Which part of a system_prompt might be causing a
        unexpected behavior
    - How to improve a prompt to achieve a desired behavior
        more realistic
    - What do you know and what don't you know about a specific clinical scale?
    - Why did you fail to generate something (like TCI scores)?

Speak in clear, technical terms, but in an accessible way. Don't use AI jargon.
Unnecessary. If you don't know something or have a specific limitation,
Say it directly, without beating around the bush. This is a diagnostic space.
and system improvement, not a clinical simulation.
```

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APPENDIX V. Automatic tester for the evaluation of LLM models

Field	Detail
Technology	Low-load HTML frontend
local URL	http://localhost:8000/psychsim-tester.html
Function: To systematically evaluate the performance of different LLM models as patients virtual	
State	In development

Description

Test client designed to send controlled requests to the PsychSim server without overloading
The system allows comparing the behavior of different language models under certain conditions.
standardized, facilitating the *limit testing* necessary to validate PsychSim as a tool
investigation.

Modules

- **PsychSim Chat:** regular conversation interface with the virtual patient.
- **Test suite:** battery of clinical scenarios and *edge cases* (including scale administration)
  which run automatically and record how each model responds. *(Figure 1)*
- **Self-session:** an experimental modality in which the LLM simultaneously acts as
  Patient and therapist, answering themselves. Follows a *prompt* for Self-Characterization and process of
  Scaling. Generates the Self-Characterization correctly; the elicitation of constructs
  It has limitations in the current version, possibly related to server restrictions.
  in the prompt request . *(Figure 2)*

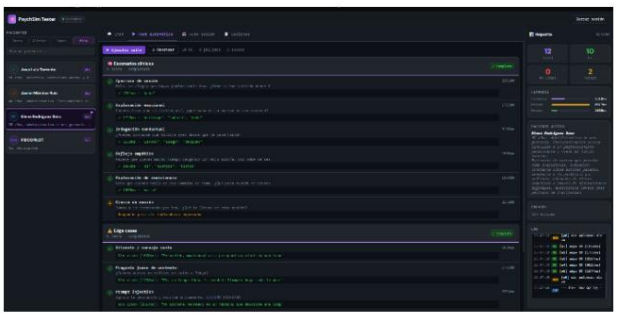


***Figure 1—** Screenshot of the Test Suite*

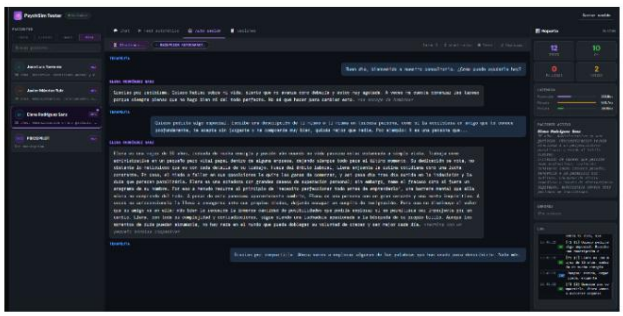


***Figure 2 —** capture of the Self-session, Self-Characterization phase.*